Unit 1. Statistical Learning

1 What Is Statistical Learning?

ISLR §2.1 ·FREES §1.1–1.6. SRM learning outcome 1(a).

1.1 The model and its two uses

Definition 1.1 (Statistical learning model)**.** For a quantitative response Y and p predictors $X = (X_1, \ldots, X_p)$,

$$Y = f(X) + \varepsilon, \tag{1}$$

where f is a fixed unknown Borel function and ε is a random error, independent of X with mean zero and finite variance $\text{var}(\varepsilon) = \mathsf{E}[\varepsilon^2] < \infty$. f is the **systematic** information X carries about Y; **statistical learning** is the set of methods for estimating f from training data $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, $x_i = (x_{i1}, \ldots, x_{ip})'$.

Remark (Terminology; FREES Table 1.3, ISLR §2.1)**.** Y is the response, output, dependent, explained or left-hand-side variable, the regressand; the X_j are predictors, inputs, features, independent, explanatory or right-hand-side variables, regressors, covariates, treatments (FREES Table

1.3, ISLR §2.1). The pairs are used interchangeably in both texts and on the examination.

Proposition 1.2 (The regression function)**.** Let $\mathsf{E}[Y^2] < \infty$, write $\mathsf{E}[Y \mid X]$ for the conditional expectation of Y given $\sigma(X)$ (the prerequisites' theorem **conditional expectation in** L^2), and let f^* be a Borel function with $f^*(X) = \mathsf{E}[Y \mid X]$ almost surely, whose existence is assumed here; $f^*(x)$ is written $\mathsf{E}[Y \mid X = x]$. For every Borel function g,

$$\mathsf{E}\big[(Y - f^*(X))^2\big] \leqslant \mathsf{E}\big[(Y - g(X))^2\big],$$

with equality iff $g(X) = f^*(X)$ almost surely. Further, $\varepsilon^* = Y - f^*(X)$ has conditional mean zero, and if Y satisfies (1) then the f there satisfies $f(X) = f^*(X)$ almost surely.

Proof. If $\mathsf{E}[g(X)^2] = \infty$ then $\mathsf{E}[(Y - g(X))^2] = \infty$, because $g(X) = Y - (Y - g(X))$ and L^2 is a vector space containing Y; the inequality is then strict, and $g(X) = f^*(X)$ almost surely fails because $f^*(X) \in L^2$ (below), so there is nothing to prove. Let now g be any Borel function with $\mathsf{E}[g(X)^2] < \infty$, so that every expectation below is finite; f^* itself qualifies, since $f^*(X) = \mathsf{E}[Y \mid X] \in L^2(\sigma(X))$ by the prerequisites' theorem. The only properties of conditional expectation used

are the orthogonality relation of that theorem with $\mathcal{G} = \sigma(X)$,

$$\mathsf{E}\big[(Y - f^*(X))\, W\big] = 0 \qquad \text{for every } W \in L^2(\sigma(X)), \tag{2}$$

and the uniqueness it asserts: a variable in $L^2(\sigma(X))$ satisfying this relation in place of $f^*(X)$ equals $f^*(X)$ almost surely.

Step 1: decompose. Put $W = f^*(X) - g(X)$, a function of X with $\mathsf{E}[W^2] < \infty$, so $W \in L^2(\sigma(X))$. Write $Y - g(X) = \{Y - f^*(X)\} + W$ and expand the square:

$$(Y - g(X))^2 = (Y - f^*(X))^2 + 2W\{Y - f^*(X)\} + W^2.$$

By (2) the middle term has expectation zero, hence

$$\mathsf{E}[(Y - g(X))^2] = \mathsf{E}[(Y - f^*(X))^2] + \mathsf{E}\big[\{f^*(X) - g(X)\}^2\big]. \tag{3}$$

Step 2: compare. The last term of (3) is non-negative, so $\mathsf{E}[(Y - f^*(X))^2] \leqslant \mathsf{E}[(Y - g(X))^2]$, with equality iff $\mathsf{E}[\{f^*(X) - g(X)\}^2] = 0$, that is iff $g(X) = f^*(X)$ almost surely.

Step 3: the error. With $\varepsilon^* = Y - f^*(X)$, (2) reads $\mathsf{E}[(\varepsilon^* - 0)W] = 0$ for every $W \in L^2(\sigma(X))$, so the zero variable satisfies the defining relation of $\mathsf{E}[\varepsilon^* \mid X]$ in the prerequisites' theorem and, by its uniqueness, $\mathsf{E}[\varepsilon^* \mid X] = 0$; $W \equiv 1$ gives $\mathsf{E}[\varepsilon^*] = 0$.

Step 4: identification. Now let ε denote the error of (1), not the ε^* of Step 3, and suppose $Y = f(X) + \varepsilon$ with ε independent of X and $\mathsf{E}[\varepsilon] = 0$. First $f(X) \in L^2$: by independence and Tonelli's theorem, $\infty > \mathsf{E}[Y^2] = \int \mathsf{E}\big[(f(x) + \varepsilon)^2\big]\, \mathsf{P}_X(dx)$, so $\mathsf{E}[(f(x_1) + \varepsilon)^2] < \infty$ for some x_1, whence $\mathsf{E}[\varepsilon^2] < \infty$; then $\mathsf{E}[(f(x) + \varepsilon)^2] = f(x)^2 + \mathsf{E}[\varepsilon^2]$ for every x, and $\mathsf{E}[f(X)^2] = \mathsf{E}[Y^2] - \mathsf{E}[\varepsilon^2] < \infty$. Next, for $W \in L^2(\sigma(X))$, W is independent of ε and εW is integrable by the Cauchy–Schwarz inequality, so $\mathsf{E}[(Y - f(X))W] = \mathsf{E}[\varepsilon W] = \mathsf{E}[\varepsilon]\ \mathsf{E}[W] = 0$. Thus $f(X) \in L^2(\sigma(X))$ satisfies (2) in place of $f^*(X)$, and by uniqueness $f(X) = \mathsf{E}[Y \mid X] = f^*(X)$ almost surely. □

Example 1.1 (Galton's 1885 data; FREES §1.1)**.** Galton classified 928 adult children by their own height and by an index of their parents' height, the average of the father's height and 1.08 times the mother's, both in inches (FREES Table 1.1). With Y the child's height and X the parents' index, the mean of Y within each class of X estimates the f^* of Proposition 1.2.

```
galton <- read.csv("data/frees/csv/Galton.csv")
sum(galton$NUMBER)
```

```
## [1] 928
```

```
class_means <- sapply(split(galton, galton$PARENTC),
                      function(s) weighted.mean(s$CHILDC, s$NUMBER))
round(class_means, 2)
```

```
##  63.5  64.5  66.5  67.5  68.5  69.5  70.5  71.5  72.5  73.5
## 65.27 65.37 66.70 67.06 67.59 68.02 68.72 69.61 70.11 71.99
##  74.5
## 72.95
```

```
c(rise_children = diff(range(class_means)),
  rise_parents  = diff(range(as.numeric(names(class_means)))))
```

```
## rise_children  rise_parents
##         7.679        11.000
```

The class means rise with the parents' index, but by 7.7 inches over its 11-inch range: the children of tall parents are tall, and their excess over the average is smaller than their parents' own, the

regression toward the mean from which the method takes its name.

Definition 1.3 (Prediction and inference)**.** In **prediction** the inputs X are available, Y is not, and $\hat{Y} = \hat{f}(X)$ is reported; $\hat{f}$ is a black box and only its accuracy matters. In **inference** the object is f itself: which predictors are associated with the response, whether each relationship is positive or negative, and whether it is adequately summarised by a linear equation.

Proposition 1.4 (Reducible and irreducible error)**.** Fix $\hat{f}$ and X; since the test pair (X, ε) is independent of the training data that produced $\hat{f}$, and ε is independent of X (Definition 1.1), ε is independent of $(X, \hat{f})$ jointly, so its law is unchanged and the only randomness is in ε. Then

$$\mathsf{E}(Y - \hat{Y})^2 = \underbrace{\big[f(X) - \hat{f}(X)\big]^2}_{\text{reducible}} + \underbrace{\operatorname{var}(\varepsilon)}_{\text{irreducible}} . \tag{4}$$

Proof. By (1), $Y - \hat{Y} = f(X) + \varepsilon - \hat{f}(X) = a + \varepsilon$ with $a = f(X) - \hat{f}(X)$, which is a constant

because $\hat{f}$ and X are fixed. Then

$$(Y - \hat{Y})^2 = (a + \varepsilon)^2 = a^2 + 2a\varepsilon + \varepsilon^2,$$

and taking expectations term by term,

$$\mathsf{E}(Y - \hat{Y})^2 = a^2 + 2a\ \mathsf{E}[\varepsilon] + \mathsf{E}[\varepsilon^2] = a^2 + 0 + \mathrm{var}(\varepsilon),$$

since $\mathsf{E}[\varepsilon] = 0$ gives $\mathsf{E}[\varepsilon^2] = \mathrm{var}(\varepsilon) + (\mathsf{E}\,\varepsilon)^2 = \mathrm{var}(\varepsilon)$. Substituting a gives (4). □

Remark (ISLR §2.1.1)**.** The reducible error can be lowered by a better estimate of f; the irreducible error cannot, because ε carries unmeasured variables and unmeasurable variation. $\mathrm{var}(\varepsilon)$ is an upper bound on prediction accuracy and is almost always unknown in practice.

Example 1.2 (Which goal; ISLR §2.1.1)**.** A direct-marketing campaign that wants only to identify likely responders is prediction. Asking which advertising media are associated with sales, and how large an increase in sales accompanies a given increase in TV advertising, is inference. Relating house values to crime rate, air quality, schools and size serves both: the premium attached to

a river view is inference, whether a given house is over- or under-valued is prediction.

1.2 Estimating f: parametric and non-parametric methods

Definition 1.5 (Parametric method)**.** A **parametric** method (i) assumes a functional form for f, for example the linear model

$$f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p, \tag{5}$$

and (ii) fits the parameters from the training data, most often by least squares. Estimating f is reduced to estimating $p + 1$ numbers.

Definition 1.6 (Non-parametric method)**.** A **non-parametric** method assumes no functional form and seeks an $\hat{f}$ as close to the training points as possible subject to a smoothness restriction, for example a thin-plate spline with a chosen level of smoothness.

Remark (The trade; ISLR §2.1.2)**.** A parametric form may be far from the true f, in which case the estimate is poor whatever the sample size. A non-parametric method avoids that risk and

needs far more observations, nothing being borrowed from an assumed shape; and its smoothness has to be chosen, since a fit that follows every training point has **overfit** the data and predicts new observations badly.

Definition 1.7 (Flexibility and interpretability; ISLR §2.1.3)**.** A method is more **flexible** the wider the range of shapes it can produce for $\hat{f}$. Ordered from least to most flexible (ISLR Figure 2.7): subset selection and the lasso; least squares; generalised additive models and trees; bagging, boosting, support vector machines and neural networks. Interpretability runs the other way: the lasso is more interpretable than least squares because it sets coefficients to zero, and the fully non-linear methods are the hardest to interpret.

Remark (ISLR §2.1.3)**.** When inference is the goal, a restrictive model is preferred for its interpretability. When prediction is the sole goal, the most flexible method is not necessarily the most accurate: flexible methods can overfit, and a less flexible method often predicts better.

1.3 Classes of problem

Definition 1.8 (Supervised and unsupervised learning)**.** Learning is **supervised** when each predictor vector x_i comes with a response y_i and the aim is prediction or inference about their relationship. It is **unsupervised** when there is no response: the aim is then to find structure among the variables or the observations, as in **cluster analysis**, which asks whether the observations fall into distinct groups.

Remark (ISLR §2.1.4)**.** Between the two lies **semi-supervised** learning: of n observations, $m < n$ have both predictors and a response and the other $n - m$ have predictors only, as when the response is expensive to measure. Its methods are beyond these notes.

Definition 1.9 (Regression and classification)**.** A variable is **quantitative** if it takes numerical values and **qualitative** (categorical) if it takes values in one of J classes (J rather than ISLR's K, which below denotes the number of neighbours in KNN). A supervised problem is a **regression** problem when the response is quantitative and a **classification** problem when it is qualitative.

Remark (ISLR §2.1.5)**.** The distinction rests on the response, not the predictors: qualitative predictors are handled by coding. Logistic regression is a classification method despite its name, since its response is a binary class, though as it estimates class probabilities it can also be viewed as regression. K-nearest neighbours and boosting serve both kinds of response.

1.4 Data, sampling and study design

Definition 1.10 (Summary statistics; FREES §1.2)**.** For a sample $y_1, \ldots, y_n$: the mean $\bar{y} = n^{-1}\sum_i y_i$; the median, the middle value when the observations are ordered (the average of the two middle values when n is even); the standard deviation $s_y = \{(n-1)^{-1}\sum_i (y_i - \bar{y})^2\}^{1/2}$; and the $100q$-th percentile, a value below which a fraction q of the observations lie. A histogram, a box plot (the box is the middle 50%, the whiskers extend to the most extreme observations within 1.5 times the box length of the box ends (R's hinges, `fivenum`, which are close to the quartiles), and points outside them are plotted individually) and a normal quantile–quantile plot (sample quantiles against standard normal quantiles; a straight line means agreement in shape) display the distribution.

Example 1.3 (Massachusetts bodily injury claims; FREES §1.2)**.** Rempala and Derrig (2005) sampled 348 automobile bodily injury claims closed in Massachusetts in 2001, the amounts of outpatient medical treatment after accidents. The claims from provider "A", suspected of fraud, are set aside, and the others are analysed in logarithms of thousands of dollars, the scale of FREES Table 1.2.

```
injury <- read.csv("data/frees/csv/MassBodilyInjury.csv")
names(injury) <- trimws(names(injury))
table(injury$provider)
```

```
##
##     A Other
##    76   272
```

```
claims  <- subset(injury, provider == "Other")$claims
lclaims <- log(claims)
c(n = length(lclaims), mean = mean(lclaims), median = median(lclaims), sd =
 ↪  sd(lclaims),
  min = min(lclaims), max = max(lclaims),
  q25 = unname(quantile(lclaims, 0.25)), q75 = unname(quantile(lclaims, 0.75)))
```

```
##         n      mean    median        sd       min       max
##  272.0000    0.4807    0.7930    1.1009   -3.1011    3.9120
##       q25       q75
##   -0.1138    1.1678
```

```
c(mean_in_dollars = 1000 * exp(mean(lclaims)),
  normal_q75 = mean(lclaims) + qnorm(0.75) * sd(lclaims))
```

```
## mean_in_dollars      normal_q75
##        1617.262           1.223
```

The mean logarithmic claim, 0.48, is \$1,617. The normal curve with $\mu = \bar{y}$ and $\sigma = s_y$ summarises the distribution by two numbers; its 75th percentile, $\bar{y} + 0.675\, s_y = 1.22$, is close to the sample's 1.17.

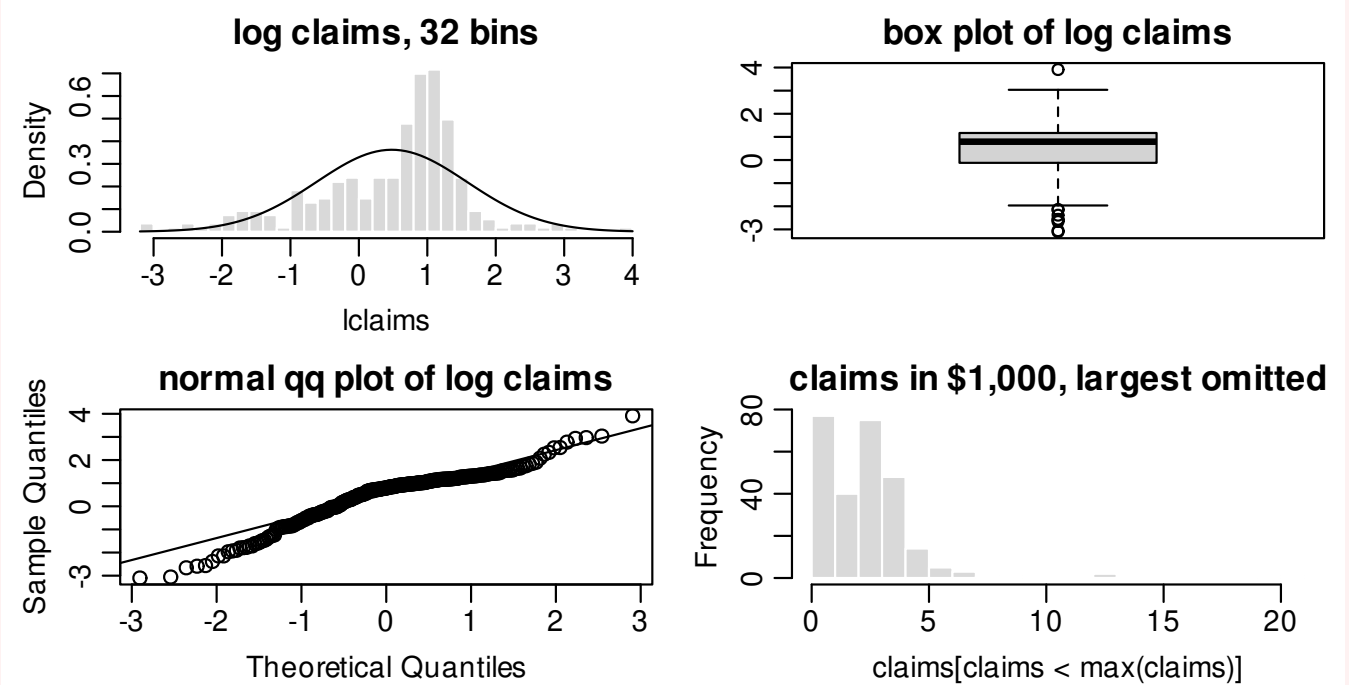


Figure 1.1: The 272 claims of the other providers: histogram of the logarithms with the fitted normal curve, box plot, normal quantile–quantile plot, and the histogram in thousands of dollars with the largest claim omitted (FREES Figures 1.3–1.6).

The quantile plot falls below the line in the lower tail and the box plot has its outliers mostly below the lower whisker: the logarithms are skewed to the left, with a point beyond the whiskers at each end. On the dollar scale the distribution is lopsided to the right.

Definition 1.11 (Skewness and power transforms; FREES §1.3)**.** A distribution with the longer tail on the right is **skewed to the right**, or positively skewed, and one with the longer tail on the left is skewed to the left; the sample skewness

$$\frac{n^{-1}\sum_i (y_i-\bar{y})^3}{\{n^{-1}\sum_i (y_i-\bar{y})^2\}^{3/2}}$$

is positive in the first case and negative in the second. Insurance claim distributions are typically skewed to the right. The Box–Cox family of **power transforms** replaces a positive response y by

$$y^{(\lambda)} = \begin{cases} (y^\lambda - 1)/\lambda, & \lambda \neq 0, \\ \ln y, & \lambda = 0, \end{cases}$$

with $\lambda = 1, \frac{1}{2}, 0, -1$ the usual choices; in practice y^λ or $\ln y$ is used directly, with the sign reversed

for $\lambda < 0$ so that large values stay large. $y^{(\lambda)} \to \ln y$ as $\lambda \to 0$, since $y^{\lambda} = e^{\lambda \ln y} = 1 + \lambda \ln y + O(\lambda^2)$.

```
skew <- function(v) mean((v - mean(v))^3) / mean((v - mean(v))^2)^1.5
c(dollars = skew(claims), log = skew(lclaims))
```

```
## dollars     log
##  7.5532 -0.7564
```

The logarithm takes the claims from a skewness of 7.6 on the dollar scale to -0.8: it compresses the upper tail of a right-skewed distribution and can overshoot into left skew.

Example 1.4 (Transforms of simulated chi-square data; FREES §1.3)**.** FREES Figure 1.7 draws 500 observations from a χ^2_2 distribution and rescales them by the square root, the logarithm and the negative reciprocal; the chapter's script fixes the seed.

```
set.seed(1237)
x1 <- 10000 * rchisq(500, df = 2)
tr <- list(original = x1, sqrt = sqrt(x1), log = log(x1), neg_reciprocal = -1 / x1)
round(sapply(tr, skew), 3)
```

```
##       original           sqrt            log neg_reciprocal
##          1.656          0.642         -1.123         -21.079
```

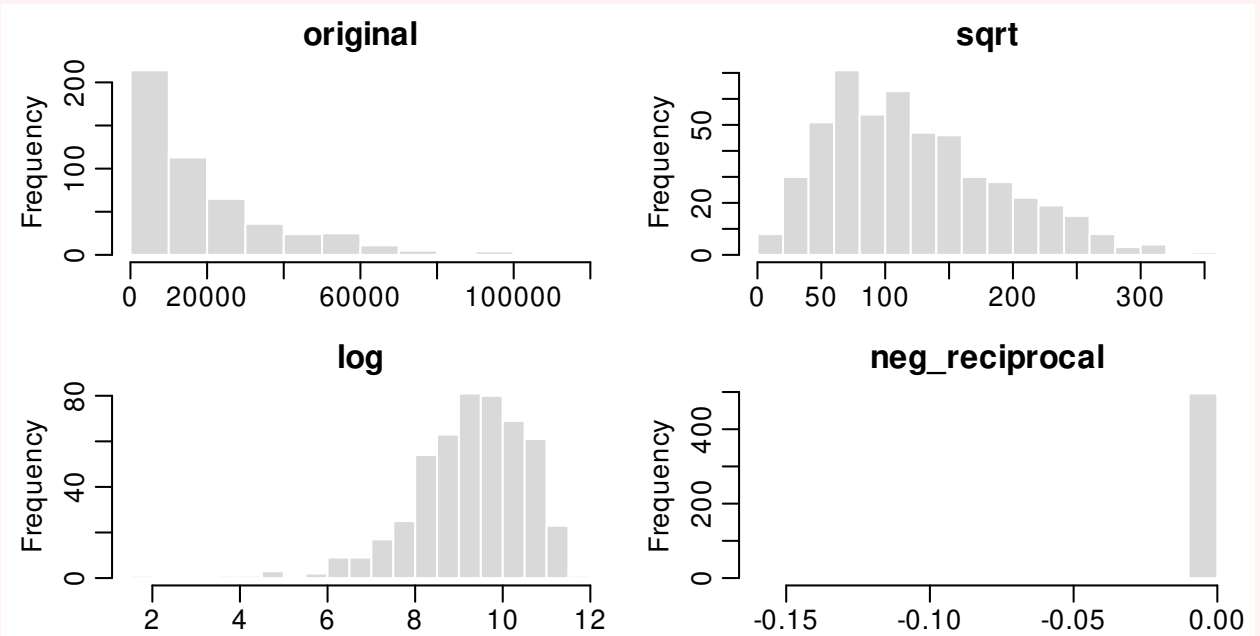


Figure 1.2: The 500 simulated χ^2_2 observations, raw and under the three transforms (FREES Figure 1.7).

The square root brings the skewness from 1.66 to 0.64 and the logarithm reverses its sign, to −1.12; by the coefficient the square root is the most nearly symmetric of the four, while FREES reads the logarithm as the best from the histograms. The negative reciprocal is dominated by the smallest draws and is far more skewed than the original.

Definition 1.12 (Basic sampling assumptions; FREES §1.4)**.** The data $\{y_i\}$ are drawn from a population with (1) $\mathsf{E}[y_i] = \mu$, (2) $\operatorname{var}(y_i) = \sigma^2$, (3) the y_i independent, (4) the y_i normally distributed. μ and σ^2 are the location and scale parameters of the population, estimated by $\bar{y}$ and s_y^2.

Remark (FREES §1.4)**.** Assumption (4) is not required for most inference, because the central limit theorem of the prerequisites makes $\bar{y}$, and the weighted averages that regression coefficients are, approximately normal; it is required for the exact justification of t statistics.

Definition 1.13 (Designed and observational studies; FREES §1.5)**.** In a **designed study** the analyst controls the data: treatments are assigned to units by **randomisation**, and a treatment effect can then be read as **causal**. In an **observational study** the data are not under the ana-

lyst's control. A **confounding variable** is one related to both the response and an explanatory variable; it can produce association without causation. **Statistical control** compares y and x "controlling for" other explanatory variables by including them in the regression, which is the essence of the regression method and the only control available in observational data.

Example 1.5 (Rand Health Insurance Experiment; FREES §1.5)**.** Medical expenditure and demand for insurance are positively associated in observational data, but people in poor health both spend more and buy more insurance. The Rand experiment randomly assigned households to fourteen plans differing in co-insurance and deductible, and found that more generous plans caused greater expenditure even after controlling for health status. Actuarial data are almost always observational: a smoking study cannot randomise people into smokers, so its output is an association adjusted for what was measured.

Remark (Actuarial applications; FREES §1.6)**.** Pricing: expected claims vary with sex, age, location and vehicle use; a classification system built by regression lets each insured pay a fair share, and a flat rate loses good risks to competitors while retaining bad ones, which is **adverse selection**. Reserving and solvency testing: regression forecasts future obligations and compares

healthy with distressed firms.

2 Assessing Model Accuracy

ISLR §2.2. SRM learning outcomes 1(b), 1(c).

2.1 Training and test error

Remark (ISLR §2.2)**.** ISLR's **no free lunch**: no one method dominates all others over all possible data sets. One method may work best on a particular data set and another on a similar but different one, so choosing the method for the data at hand is part of the analysis, and the criteria below are the tools for that choice.

Definition 1.14 (Mean squared error)**.** For a fit $\hat{f}$ obtained from the training data,

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\big(y_i - \hat{f}(x_i)\big)^2 \tag{6}$$

is the **training MSE**. The **test MSE** is the average $\text{Ave}\,(y_0 - \hat{f}(x_0))^2$ over previously unseen

test observations (x_0, y_0) not used to train the method.

Remark (ISLR §2.2.1)**.** The method with the lowest training MSE need not have the lowest test MSE: most methods choose $\hat{f}$ to make the training MSE small, so it tends to understate the test MSE. For nested least-squares classes the training MSE cannot increase with flexibility, each fit minimising over a larger set; in general it tends to decrease. The test MSE typically falls and then rises, a **U-shape**, but need not. A method with small training MSE and large test MSE is **overfitting**: it has followed patterns caused by chance in the training data that are absent from the test data. Overfitting refers to the case in which a less flexible model would have given a smaller test MSE. ISLR measures flexibility by the **degrees of freedom** of a fit, a count that summarises how flexible a curve is: for a least-squares fit it is the number of fitted coefficients, the polynomial degree plus one in the simulation below.

```r
f     <- function(x) sin(2 * x) + 0.5 * x      # a known truth
sigma <- 0.5
```

```r
set.seed(4)
dtr <- data.frame(x = runif(80, -3, 3));    dtr$y <- f(dtr$x) + rnorm(80, sd = sigma)
dte <- data.frame(x = runif(20000, -3, 3)); dte$y <- f(dte$x) + rnorm(20000, sd =
 ↪  sigma)
degs <- 1:16
tt <- t(sapply(degs, function(dg) {
  m <- lm(y ~ poly(x, dg), data = dtr)
  c(train = mean(residuals(m)^2),
    test  = mean((dte$y - predict(m, newdata = dte))^2))
}))
round(t(tt), 3)[, c(1, 2, 4, 8, 12, 16)]
```

```
##        [,1]  [,2]  [,3]  [,4]  [,5]  [,6]
## train 0.616 0.603 0.482 0.207 0.202 0.184
## test  0.684 0.698 0.579 0.259 0.263 0.325
```

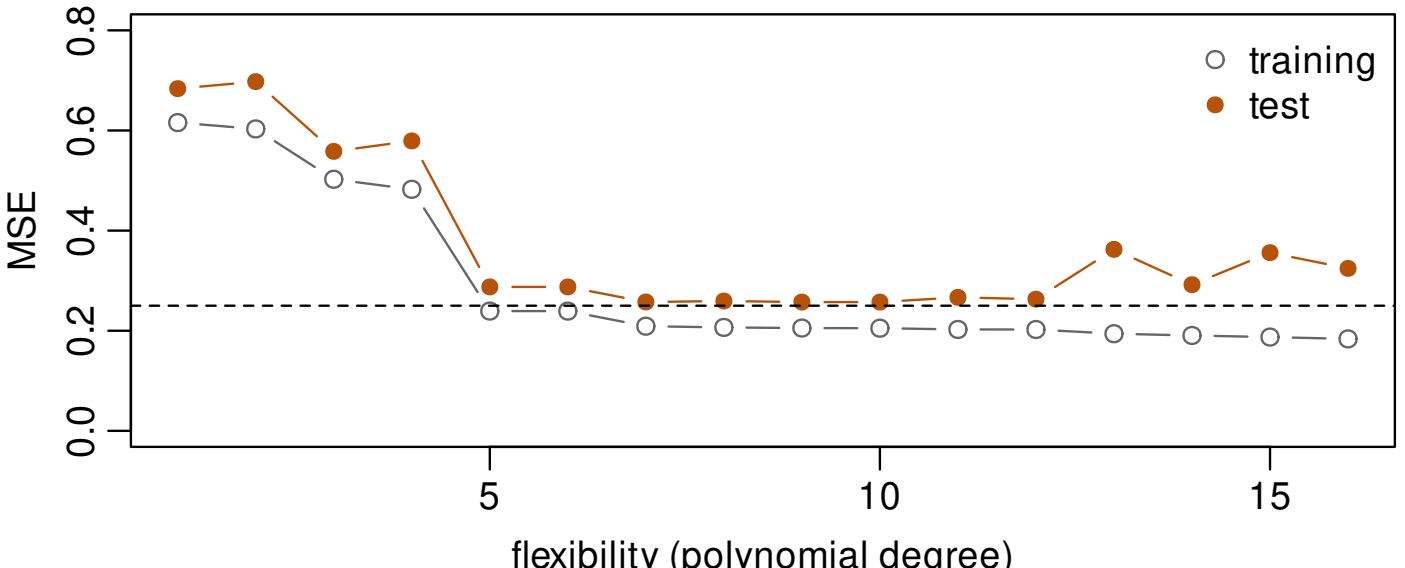


Figure 1.3: Training and test MSE for polynomial fits of increasing degree to 80 observations from $y = \sin 2x + x/2 + \varepsilon$, $\varepsilon \sim N(0, 0.5^2)$. The dashed line is $\mathrm{var}(\varepsilon)$, the irreducible error and the lowest achievable test MSE.

2.2 The bias–variance trade-off

Proposition 1.15 (Bias–variance decomposition)**.** At a fixed test point x_0 at which the estimate has $\mathsf{E}[\hat{f}(x_0)^2] < \infty$, the expected test MSE over repeated training sets and a new y_0 is

$$\mathsf{E}\big(y_0 - \hat{f}(x_0)\big)^2 = \operatorname{var}\big(\hat{f}(x_0)\big) + \big[\operatorname{Bias}\big(\hat{f}(x_0)\big)\big]^2 + \operatorname{var}(\varepsilon), \tag{7}$$

where $\operatorname{var}(\hat{f}(x_0))$ is the variance of the estimate over training sets and $\operatorname{Bias}(\hat{f}(x_0)) = \mathsf{E}[\hat{f}(x_0)] - f(x_0)$. The overall expected test MSE averages (7) over the test points x_0.

Proof. Let $\mathcal{D}$ denote the training set, $\hat{f} = \hat{f}_{\mathcal{D}}$ the estimate fitted on it, and $y_0 = f(x_0) + \varepsilon_0$ a new response at the fixed point x_0, with ε_0 independent of $\mathcal{D}$, $\mathsf{E}[\varepsilon_0] = 0$ and $\operatorname{var}(\varepsilon_0) = \operatorname{var}(\varepsilon)$. Write $m = \mathsf{E}_{\mathcal{D}}[\hat{f}(x_0)]$, a number. Then

$$y_0 - \hat{f}(x_0) = \underbrace{\varepsilon_0}_{A} + \underbrace{\{f(x_0) - m\}}_{B} + \underbrace{\{m - \hat{f}(x_0)\}}_{C},$$

where A has mean zero, $B = -\operatorname{Bias}(\hat{f}(x_0))$ is a constant, and C depends on $\mathcal{D}$ alone with

$\mathsf{E}[C] = m - \mathsf{E}_{\mathcal{D}}[\hat{f}(x_0)] = 0$. Squaring,

$$(A + B + C)^2 = A^2 + B^2 + C^2 + 2AB + 2AC + 2BC.$$

Take expectations term by term. $\mathsf{E}[A^2] = \mathrm{var}(\varepsilon_0) = \mathrm{var}(\varepsilon)$; $B^2 = [\mathrm{Bias}(\hat{f}(x_0))]^2$; $\mathsf{E}[C^2] = \mathsf{E}_{\mathcal{D}}[(\hat{f}(x_0) - m)^2] = \mathrm{var}(\hat{f}(x_0))$. The cross terms vanish: $\mathsf{E}[AB] = B\ \mathsf{E}[A] = 0$ and $\mathsf{E}[BC] = B\ \mathsf{E}[C] = 0$ because B is a constant, and $\mathsf{E}[AC] = \mathsf{E}[A]\ \mathsf{E}[C] = 0$ because A is a function of ε_0, C a function of $\mathcal{D}$, and the two are independent. Adding the six terms gives (7). The overall expected test MSE follows by averaging the identity over the distribution of x_0. □

Remark (ISLR §2.2.2)**. Variance** is the amount by which $\hat{f}$ would change if estimated from a different training set; flexible methods have high variance. **Bias** is the error from approximating a complicated f by a simpler model; flexible methods have low bias. As flexibility increases the bias at first falls faster than the variance rises and the test MSE falls; beyond some point the bias has little further to fall while the variance keeps rising, and the test MSE rises. Both terms are non-negative, so the expected test MSE never lies below $\mathrm{var}(\varepsilon)$. If the true f is linear, least squares

has no bias and a flexible method cannot compete; if f is highly non-linear and n is large, the flexible method wins.

```r
set.seed(5)
x0 <- 1.0; B <- 1000; nsim <- 60
bv <- t(sapply(c(1, 2, 4, 6, 8, 12), function(dg) {
  fh <- y0 <- numeric(B)
  for (b in seq_len(B)) {
    d <- data.frame(x = runif(nsim, -3, 3)); d$y <- f(d$x) + rnorm(nsim, sd = sigma)
    fh[b] <- predict(lm(y ~ poly(x, dg), data = d), newdata = data.frame(x = x0))
    y0[b] <- f(x0) + rnorm(1, sd = sigma)
  }
  c(degree = dg, bias2 = (mean(fh) - f(x0))^2, variance = var(fh),
    irreducible = sigma^2, sum = (mean(fh) - f(x0))^2 + var(fh) + sigma^2,
    direct = mean((y0 - fh)^2))
}))
round(bv, 4)
```

```
##      degree  bias2 variance irreducible    sum direct
## [1,]      1 1.1763   0.0141        0.25 1.4404 1.4532
## [2,]      2 1.1534   0.0259        0.25 1.4293 1.4145
```

```
## [3,]      4 0.4294   0.0347        0.25 0.7141 0.7395
## [4,]      6 0.0078   0.0267        0.25 0.2845 0.2943
## [5,]      8 0.0001   0.0338        0.25 0.2839 0.2865
## [6,]     12 0.0001   0.0466        0.25 0.2967 0.3160
```

`sum` and `direct` agree to Monte-Carlo error at every degree, which is (7); the squared bias falls to zero by degree 8 while the variance is smallest at degree 1 and largest at degree 12, and their sum is smallest in between.

2.3 The classification setting

Definition 1.16 (Error rate)**.** For a qualitative response, the **training error rate** is

$$\frac{1}{n}\sum_{i=1}^{n} I(y_i \neq \hat{y}_i), \tag{8}$$

with $\hat{y}_i$ the predicted class and I the indicator; the **test error rate** is Ave $I(y_0 \neq \hat{y}_0)$ over test observations. A good classifier has the smallest test error rate.

Definition 1.17 (Bayes classifier)**.** The **Bayes classifier** assigns a test observation x_0 to the class j for which the conditional probability $\mathsf{P}(Y = j \mid X = x_0)$ is largest ($\mathsf{P}(Y = j \mid X = x)$ is the $\mathsf{E}[I(Y = j) \mid X = x]$ of the convention of Proposition 1.2, applied to $I(Y = j)$, whose existence is assumed there). With two classes it predicts class one when $\mathsf{P}(Y = 1 \mid X = x_0) > 0.5$; the set where the probability equals 0.5 is the **Bayes decision boundary**. Its error rate at x_0 is $1 - \max_j \mathsf{P}(Y = j \mid X = x_0)$ and overall

$$1 - \mathsf{E}\Big[\max_j \mathsf{P}(Y = j \mid X)\Big], \tag{9}$$

the **Bayes error rate**, the classification analogue of the irreducible error.

Proposition 1.18 (Optimality of the Bayes classifier)**.** Let g^* be the Bayes classifier. For every classifier g, $\mathsf{P}(g^*(X) \neq Y) \leqslant \mathsf{P}(g(X) \neq Y)$: the Bayes error rate (9) is the lowest expected test error rate (misclassification probability $\mathsf{P}(g(X) \neq Y)$) any classifier can achieve.

Proof. For $j = 1, \dots, J$ write $p_j(X) = \mathsf{E}[I(Y = j) \mid X]$, the conditional expectation given $\sigma(X)$ of the prerequisites' theorem **conditional expectation in** L^2, with p_j a Borel function of x whose existence is assumed as in Proposition 1.2; $p_j(x)$ is the $\mathsf{P}(Y = j \mid X = x)$ of Definition 1.17, and $g^*(x)$ is the smallest index attaining $\max_j p_j(x)$ (any measurable choice will do), a Borel function of x. A classifier is a Borel function g of x with values in $\{1, \dots, J\}$. The theorem gives, for each j, the orthogonality relation $\mathsf{E}[\{I(Y = j) - p_j(X)\}W] = 0$ for every $W \in L^2(\sigma(X))$.

Step 1: the probability of a correct classification. Exactly one of the indicators $I(g(X) = j)$ equals one, so $I(g(X) = Y) = \sum_j I(g(X) = j)\, I(Y = j)$. Each $W_j = I(g(X) = j)$ is bounded and $\sigma(X)$-measurable, so the orthogonality relation with $W = W_j$ gives $\mathsf{E}[W_j I(Y = j)] = \mathsf{E}[W_j p_j(X)]$. Summing over j,

$$\mathsf{P}(g(X) = Y) = \mathsf{E}\Big[\sum_j I(g(X) = j)\, p_j(X)\Big] = \mathsf{E}\big[p_{g(X)}(X)\big].$$

Step 2: pointwise comparison. Since $p_{g(x)}(x)$ is one of the numbers $p_1(x), \dots, p_J(x)$ and

$p_{g^*(x)}(x)$ is the largest of them,

$$p_{g(x)}(x) \;\leqslant\; p_{g^*(x)}(x) = \max_j p_j(x) \qquad \text{for every } x.$$

Step 3: integrate. Expectation preserves the pointwise inequality of Step 2, so by Step 1 for g and for g^*, $\mathsf{P}(g(X) = Y) = \mathsf{E}[p_{g(X)}(X)] \leqslant \mathsf{E}[\max_j p_j(X)] = \mathsf{P}(g^*(X) = Y)$. Taking complements,

$$\mathsf{P}(g^*(X) \neq Y) = \mathsf{E}\Big[1 - \max_j p_j(X)\Big] \;\leqslant\; \mathsf{P}(g(X) \neq Y).$$

The left side is (9). In the two-class case $I(Y = 1) + I(Y = 2) = 1$; adding the two orthogonality relations shows that $p_1(X)+p_2(X)$ satisfies the relation of the constant 1, which the constant itself also satisfies, so $p_2(X) = 1 - p_1(X)$ almost surely by uniqueness. Then $\max\{p_1, 1 - p_1\}$ is p_1 iff $p_1 \geqslant \frac{1}{2}$; at $p_1 = \frac{1}{2}$ both labels attain the maximum, so the strict threshold rule of Definition 1.17 is one Bayes classifier. Finally $1 - \max\{p_1, 1 - p_1\} = \min\{p_1, 1 - p_1\}$. □

Definition 1.19 (K-nearest neighbours)**.** Given K and a test point x_0, let $\mathcal{N}_0$ be the K training points nearest x_0. The KNN classifier estimates

$$\hat{\mathsf{P}}(Y = j \mid X = x_0) = \frac{1}{|\mathcal{N}_0|} \sum_{i \in \mathcal{N}_0} I(y_i = j) \tag{10}$$

of the conditional probability of Definition 1.17 and assigns x_0 to the class with the largest estimate. Ties, among equidistant training points in forming $\mathcal{N}_0$ and among classes with equal votes, need a rule; by default R's `class::knn` includes in $\mathcal{N}_0$ every point tied with the K-th nearest (so $|\mathcal{N}_0|$, which is K in the absence of such ties, can exceed K) and breaks ties in the vote at random.

Remark (ISLR §2.2.3)**.** The Bayes classifier is an unattainable standard, since the conditional distribution of Y given X is unknown; KNN estimates it locally and then applies the Bayes rule. K controls flexibility through $1/K$: $K = 1$ gives an overly flexible boundary with zero training error, low bias and high variance; large K gives a nearly linear boundary with low variance and high bias. The training error rate tends to fall as $1/K$ grows and the test error rate is typically U-shaped, so the training error cannot choose K.

```r
eta <- function(x) plogis(2 * x)                # true P(Y = 1 | X = x)
set.seed(3)
ntr <- 200; nte <- 20000
tr <- data.frame(x = runif(ntr, -3, 3)); tr$y <- rbinom(ntr, 1, eta(tr$x))
te <- data.frame(x = runif(nte, -3, 3)); te$y <- rbinom(nte, 1, eta(te$x))
bayes_exp <- mean(pmin(eta(te$x), 1 - eta(te$x)))  # expected Bayes error rate
bayes <- mean((eta(te$x) > 0.5) != te$y)           # realised Bayes error rate
Ks  <- c(1, 3, 5, 15, 45, 105)
err <- t(sapply(Ks, function(K) c(K = K,
  train = mean(class::knn(tr["x"], tr["x"], tr$y, k = K) != tr$y),
  test  = mean(class::knn(tr["x"], te["x"], tr$y, k = K) != te$y))))
cbind(round(err, 4), bayes = round(bayes, 4), bayes_exp = round(bayes_exp, 4))
```

```
##         K train   test  bayes bayes_exp
## [1,]    1 0.000 0.1703 0.1129     0.116
## [2,]    3 0.090 0.1284 0.1129     0.116
## [3,]    5 0.125 0.1291 0.1129     0.116
## [4,]   15 0.130 0.1130 0.1129     0.116
## [5,]   45 0.130 0.1129 0.1129     0.116
## [6,]  105 0.135 0.1136 0.1129     0.116
```

The `test` and `bayes` columns are realised error rates on the same 20,000 random test labels, each with standard error about 0.002; `bayes_exp` is the expected rate (9) given the test x's. Proposition 1.18 orders expected rates, so a realised KNN rate may fall slightly below `bayes_exp`.

3 Resampling Methods

ISLR §5.1. SRM learning outcome 1(d).

3.1 The validation set approach

Definition 1.20 (Validation set approach)**.** Randomly divide the observations into a **training set** and a **validation set** (hold-out set). Fit on the training set; the MSE of the fitted model on the validation set estimates the test MSE.

Remark (Two drawbacks; ISLR §5.1.1)**.** (1) The estimate is highly variable: a different random split gives a different estimate, and different splits may disagree about which model is best. (2) Only the training subset, typically half the data, is used to fit, and methods perform worse with fewer observations, so the validation error tends to **overestimate** the test error of the model

fitted to all the data.

3.2 Leave-one-out cross-validation

Definition 1.21 (LOOCV)**.** For $i = 1, \dots, n$, fit on the $n-1$ observations other than (x_i, y_i), predict $\hat{y}_i^{(-i)}$ for the excluded one, and set $\mathrm{MSE}_i = (y_i - \hat{y}_i^{(-i)})^2$. The LOOCV estimate is

$$\mathrm{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{MSE}_i. \tag{11}$$

Remark (ISLR §5.1.2)**.** When the n observations are independent draws from the law of a test observation, each MSE_i is exactly unbiased for the test error of the method trained on $n-1$ observations, since (x_i, y_i) is then independent of that fit and distributed as a test observation, and highly variable, being one squared error; the average has far less bias than the validation set estimate, each fit using $n-1$ rather than $n/2$ observations; and there is no randomness in the splits, so repeating LOOCV gives the same number. Its cost is n fits.

Definition 1.22 (Least squares and leverage)**.** For a linear or polynomial regression with $n \times (p+1)$ design matrix $\mathbf{X}$ whose i-th row is $\mathbf{x}_i'$, of full column rank $p+1$ so that $\mathbf{X}'\mathbf{X}$ is invertible (for a polynomial fit of degree d in one predictor, at least $d+1$ distinct x values), least squares gives $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, fitted values $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, residuals $e_i = y_i - \hat{y}_i$, and the **leverage** of observation i is $h_i = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$, the i-th diagonal entry of $\mathbf{H}$.

Lemma 1.23 (Leverage bounds)**.** $0 \leqslant h_i \leqslant 1$, and $\partial \hat{y}_i / \partial y_i = h_i$.

Proof. $\mathbf{H}$ is symmetric, $\mathbf{H}' = \mathbf{X}[(\mathbf{X}'\mathbf{X})^{-1}]'\mathbf{X}' = \mathbf{H}$ because $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric, and idempotent, $\mathbf{H}^2 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{H}$. Comparing the (i,i) entries of $\mathbf{H} = \mathbf{H}^2 = \mathbf{H}'\mathbf{H}$,

$$h_i = \sum_{j=1}^{n} h_{ji} h_{ji} = \sum_{j=1}^{n} h_{ij}^2 = h_i^2 + \sum_{j \neq i} h_{ij}^2.$$

The middle expression is a sum of squares, so $h_i \geqslant 0$; dropping the non-negative tail of the right side gives $h_i \geqslant h_i^2$, that is $h_i(1 - h_i) \geqslant 0$, so $h_i \leqslant 1$. Finally $\hat{y}_i = \sum_j h_{ij} y_j$, whose derivative in y_i is $h_{ii} = h_i$. □

Proposition 1.24 (LOOCV shortcut for least squares)**.** For a linear or polynomial regression fitted by least squares,

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2, \tag{12}$$

where $\hat{y}_i$ and h_i are the fitted value and leverage of observation i in the single fit on all n observations, provided $h_i < 1$ for every i. The cost of LOOCV is then that of one fit.

Proof. Step 1: a rank-one inverse update (Sherman–Morrison). Let $\mathbf{A}$ be invertible and $\mathbf{u}$ a vector with $c := 1 - \mathbf{u}'\mathbf{A}^{-1}\mathbf{u} \neq 0$. Then

$$(\mathbf{A} - \mathbf{u}\mathbf{u}')^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'\mathbf{A}^{-1}}{c}. \tag{13}$$

To verify, multiply the right side by $\mathbf{A}-\mathbf{u}\mathbf{u}'$ on the right and use that $\mathbf{u}'\mathbf{A}^{-1}\mathbf{u}$ is a scalar:

$$\begin{aligned}\left(\mathbf{A}^{-1}+\tfrac{1}{c}\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'\mathbf{A}^{-1}\right)(\mathbf{A}-\mathbf{u}\mathbf{u}') &= \mathbf{I}-\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'+\tfrac{1}{c}\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'-\tfrac{1}{c}\mathbf{A}^{-1}\mathbf{u}\,(\mathbf{u}'\mathbf{A}^{-1}\mathbf{u})\,\mathbf{u}' \\ &= \mathbf{I}-\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'+\tfrac{1}{c}\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'\,(1-\mathbf{u}'\mathbf{A}^{-1}\mathbf{u}) \\ &= \mathbf{I}-\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'+\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'=\mathbf{I}.\end{aligned}$$

This exhibits a left inverse $\mathbf{B}$ of the square matrix $\mathbf{C}=\mathbf{A}-\mathbf{u}\mathbf{u}'$. It is the inverse: $\mathbf{C}\mathbf{x}=\mathbf{0}$ gives $\mathbf{x}=\mathbf{B}\mathbf{C}\mathbf{x}=\mathbf{0}$, so $\mathbf{C}$ has nullity zero and, by the prerequisites' rank–nullity theorem, full rank; hence $\mathbf{C}^{-1}$ exists and $\mathbf{B}=\mathbf{B}\mathbf{C}\mathbf{C}^{-1}=\mathbf{C}^{-1}$.

Step 2: the deleted estimate. Deleting observation i removes the term $\mathbf{x}_i\mathbf{x}_i'$ from $\mathbf{X}'\mathbf{X}=\sum_j \mathbf{x}_j\mathbf{x}_j'$ and the term $\mathbf{x}_i y_i$ from $\mathbf{X}'\mathbf{y}=\sum_j \mathbf{x}_j y_j$, so $\hat{\boldsymbol{\beta}}^{(-i)}=(\mathbf{X}'\mathbf{X}-\mathbf{x}_i\mathbf{x}_i')^{-1}(\mathbf{X}'\mathbf{y}-\mathbf{x}_i y_i)$. Apply (13)

with $\mathbf{A} = \mathbf{X}'\mathbf{X}$ and $\mathbf{u} = \mathbf{x}_i$, for which $\mathbf{u}'\mathbf{A}^{-1}\mathbf{u} = h_i$ and $c = 1 - h_i > 0$:

$$\begin{aligned}
\hat{\boldsymbol{\beta}}^{(-i)} &= \left[(\mathbf{X}'\mathbf{X})^{-1} + \frac{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}}{1 - h_i}\right](\mathbf{X}'\mathbf{y} - \mathbf{x}_i y_i) \\
&= \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i y_i + \frac{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\left[\mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} - \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\, y_i\right]}{1 - h_i} \\
&= \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i y_i + \frac{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i(\hat{y}_i - h_i y_i)}{1 - h_i},
\end{aligned}$$

using $\mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{x}_i'\hat{\boldsymbol{\beta}} = \hat{y}_i$. Putting the two terms in $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$ over the common denominator $1 - h_i$,

$$-y_i + \frac{\hat{y}_i - h_i y_i}{1 - h_i} = \frac{-y_i(1 - h_i) + \hat{y}_i - h_i y_i}{1 - h_i} = \frac{-y_i + \hat{y}_i}{1 - h_i},$$

so

$$\hat{\boldsymbol{\beta}}^{(-i)} = \hat{\boldsymbol{\beta}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i \frac{y_i - \hat{y}_i}{1 - h_i}. \tag{14}$$

Step 3: the deleted prediction. Premultiply (14) by $\mathbf{x}_i'$ and use $\mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i = h_i$:

$$\hat{y}_i^{(-i)} = \mathbf{x}_i'\hat{\boldsymbol{\beta}}^{(-i)} = \hat{y}_i - h_i\frac{y_i - \hat{y}_i}{1-h_i}, \qquad \text{so} \qquad y_i - \hat{y}_i^{(-i)} = (y_i - \hat{y}_i)\left(1 + \frac{h_i}{1-h_i}\right) = \frac{y_i - \hat{y}_i}{1-h_i}.$$

Step 4: average. $\mathrm{MSE}_i = (y_i - \hat{y}_i^{(-i)})^2 = \{(y_i - \hat{y}_i)/(1-h_i)\}^2$; averaging over i in (11) gives (12). □

Remark (ISLR §5.1.2)**.** By Lemma 1.23 the leverage measures how much an observation influences its own fit, and the formula inflates the residual of a high-leverage point by exactly the amount that makes the identity hold; since $1-h_i \leqslant 1$, $\mathrm{CV}_{(n)} \geqslant$ MSE term by term. The shortcut does not hold in general; for logistic regression, discriminant analysis or a tree the n fits must be performed.

3.3 k-fold cross-validation

Definition 1.25 (k-fold CV)**.** Randomly divide the observations into k folds of approximately equal size. For each fold in turn, fit on the other $k-1$ folds and compute MSE_j on the held-out fold. Then

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{j=1}^{k} \mathrm{MSE}_j. \tag{15}$$

LOOCV is the case $k = n$; in practice $k = 5$ or $k = 10$.

Example 1.6 (CV against the true test MSE; ISLR Figure 5.6)**.** For the polynomial fits of §2.1, whose test MSE was computed there on 20,000 fresh observations, LOOCV by the shortcut formula and 10-fold CV on the 80 training observations alone:

```
set.seed(1); fold10 <- sample(rep(1:10, length.out = 80))
t(sapply(1:10, function(dg) { m <- lm(y ~ poly(x, dg), data = dtr)
  e <- numeric(80)
  for (j in 1:10) e[fold10 == j] <- dtr$y[fold10 == j] -
    predict(lm(y ~ poly(x, dg), data = dtr[fold10 != j, ]), dtr[fold10 == j, ])
```

```
  round(c(degree = dg, loocv = mean((residuals(m) / (1 - hatvalues(m)))^2),
          cv10 = mean(e^2), test = unname(tt[dg, "test"])), 3) }))
```

```
##       degree loocv  cv10  test
##  [1,]      1 0.643 0.631 0.684
##  [2,]      2 0.646 0.638 0.698
##  [3,]      3 0.565 0.567 0.558
##  [4,]      4 0.580 0.577 0.579
##  [5,]      5 0.279 0.262 0.288
##  [6,]      6 0.286 0.275 0.288
##  [7,]      7 0.271 0.244 0.257
##  [8,]      8 0.282 0.265 0.259
##  [9,]      9 0.300 0.289 0.257
## [10,]     10 0.423 0.369 0.257
```

Both CV curves follow the test MSE: they drop sharply at degree 5, where the polynomial first follows the sine, and both are smallest at degree 7, whose test MSE, 0.257, equals the minimum over the ten degrees to three decimals. They understate the level of the test MSE at some degrees and overstate it at others, and they rise at degree 10, where the test MSE does not; what CV is asked to do, locate the minimum, it does here, as in ISLR's Figure 5.6.

Proposition 1.26 (Mean of correlated estimates)**.** If $Z_1, \dots, Z_k$ have common variance $\sigma^2 > 0$ and common pairwise correlation $\rho \geqslant 0$, then

$$\operatorname{var}\left(\frac{1}{k}\sum_{j=1}^{k} Z_j\right) = \rho\sigma^2 + \frac{1-\rho}{k}\sigma^2,$$

which for $\rho < 1$ decreases strictly in k to the floor $\rho\sigma^2$ (at $\rho = 1$ it is constant) and for $k \geqslant 2$ increases strictly in ρ (at $k = 1$ it is constant).

Proof. By bilinearity of the covariance,

$$\operatorname{var}\left(\sum_{j=1}^{k} Z_j\right) = \sum_{j=1}^{k} \operatorname{var}(Z_j) + \sum_{j \neq l} \operatorname{cov}(Z_j, Z_l) = k\sigma^2 + k(k-1)\rho\sigma^2,$$

there being k variances equal to σ^2 and $k(k-1)$ ordered pairs $j \neq l$ with covariance $\rho\sigma^2$. Dividing

by k^2,

$$\operatorname{var}\Big(\frac{1}{k}\sum_j Z_j\Big) = \frac{\sigma^2}{k} + \frac{k-1}{k}\rho\sigma^2 = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{k}.$$

For $\rho < 1$ the second term is positive and decreasing in k, with limit 0, so the whole decreases to the floor $\rho\sigma^2$; and the expression equals $\sigma^2\{1/k + (1-1/k)\rho\}$, whose coefficient of ρ is $\sigma^2(1-1/k)$, positive for $k \geqslant 2$ and zero for $k = 1$, so for $k \geqslant 2$ it increases strictly in ρ. □

Remark (Choosing k; ISLR §5.1.3–5.1.4)**.** Computationally, k-fold needs k fits instead of n. There is also a bias–variance trade-off. Bias: LOOCV trains on $n-1$ observations and is nearly unbiased; k-fold trains on $(k-1)n/k$ and has intermediate bias; the validation set approach has the most. Variance: ISLR argues that LOOCV has the higher variance. Its n fits use almost identical training sets, so their MSE_i are highly positively correlated, while the k fits of k-fold overlap less, and by Proposition 1.26 a larger ρ raises the variance of an average when σ^2 and k are held fixed. The two estimators average quantities with different σ^2 and different k, so this is a heuristic, not a consequence of the proposition; for least squares the ordering can reverse. On ISLR's account LOOCV has lower bias but higher variance than k-fold with $k < n$, and $k = 5$ or

10 is the empirical compromise. Split-to-split variability of k-fold is much smaller than that of the validation set approach.

Remark (What cross-validation is used for; ISLR §5.1.3)**.** Sometimes the estimate of the test MSE itself is wanted. More often only the **location of the minimum** of the estimated curve matters, because cross-validation is being used to choose among methods or among levels of flexibility. The CV curves may under- or overestimate the true test MSE and still locate its minimum well.

Definition 1.27 (Cross-validation for classification)**.** Replace MSE_i by $\mathrm{Err}_i = I(y_i \neq \hat{y}_i^{(-i)})$, with $\hat{y}_i^{(-i)}$ the class predicted for the excluded observation as in Definition 1.21:

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Err}_i, \tag{16}$$

and analogously for k-fold and the validation set approach.

Example 1.7 (Choosing flexibility by CV; ISLR §5.1.5)**.** ISLR Figure 5.8 reports, for its two-class simulated data, that logistic regressions with polynomial terms of degrees 1 to 10 have test error rates U-shaped in the degree with minimum at degree 3, and that the 10-fold CV error underestimates them but attains its minimum at degree 4, where the true test error is nearly at its minimum; and that for KNN the CV curve likewise locates a good K while the training error rate declines in $1/K$ throughout. On the two-class simulation of §2.3, with one set of 10 folds for every method, logistic regressions on polynomials in x of degree 1 to 5 and KNN:

```
cost <- function(y, p) mean((p > 0.5) != y)
set.seed(1); fold <- sample(rep(1:10, length.out = ntr))
t(sapply(1:5, function(d) {
  cvp <- numeric(ntr)
  for (j in 1:10) cvp[fold == j] <- predict(glm(y ~ poly(x, d), binomial, tr[fold !=
   ↪  j, ]),
                                            tr[fold == j, ], type = "response")
  m <- glm(y ~ poly(x, d), binomial, tr)
  round(c(degree = d, train = cost(tr$y, fitted(m)), cv = cost(tr$y, cvp),
          test = cost(te$y, predict(m, te, type = "response"))), 4) }))
```

```
##      degree train    cv   test
## [1,]      1 0.130 0.125 0.1133
## [2,]      2 0.125 0.140 0.1130
## [3,]      3 0.135 0.140 0.1141
## [4,]      4 0.135 0.140 0.1140
## [5,]      5 0.125 0.140 0.1127
```

```
t(sapply(Ks, function(K) { cvp <- integer(ntr)
  for (j in 1:10) cvp[fold == j] <- as.integer(as.character(class::knn(
    tr[fold != j, "x", drop = FALSE], tr[fold == j, "x", drop = FALSE], tr$y[fold
    ↪  != j], k = K)))
  round(c(K = K, cv = mean(cvp != tr$y),
          test = mean(class::knn(tr["x"], te["x"], tr$y, k = K) != te$y)), 4) }))
```

```
##         K    cv   test
## [1,]    1 0.215 0.1703
## [2,]    3 0.190 0.1284
## [3,]    5 0.165 0.1291
## [4,]   15 0.135 0.1130
## [5,]   45 0.145 0.1129
## [6,] 105 0.145 0.1136
```

The true log odds are linear in x, and the CV error rate is smallest at degree 1, 0.125; the test error rates of all five degrees lie within 0.0014 of each other, so on these data the choice of degree matters little. For KNN the CV error rate is smallest at $K = 15$, 0.135, where the test error rate is within 0.0001 of its minimum over the six values, while $K = 1$, with training error 0, has the largest CV and test error rates. CV, computed on the 200 training observations alone, chooses well; its values are noisier than the test rates, which use 20,000 observations.

3.4 The bootstrap (beyond the syllabus)

ISLR §5.2. Not on the SRM or PA reading lists; kept because bagging (ISLR §8.2) is built on it.

Definition 1.28 (Bootstrap sample and bootstrap standard error; ISLR §5.2)**.** Given data $Z = \{z_1, \ldots, z_n\}$, a **bootstrap sample** Z^* consists of n observations drawn from Z independently and uniformly **with replacement**, so an observation may appear several times or not at all. For a statistic $\hat{\alpha}$ computed from Z, let $\hat{\alpha}^{*b}$ be its value on the b-th of B bootstrap samples. The

bootstrap standard error is

$$\mathrm{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\alpha}^{*b} - \frac{1}{B}\sum_{c=1}^{B}\hat{\alpha}^{*c}\right)^2}, \tag{17}$$

which estimates the standard deviation of $\hat{\alpha}$ over repeated samples from the population without any formula for it.

Proposition 1.29 (Out-of-bag fraction)**.** The probability that a given observation does not appear in a bootstrap sample is $q_n = (1 - 1/n)^n$, which increases in n to $e^{-1} = 0.3679$.

Proof. Step 1: the probability. Each of the n draws is uniform on $\{1, \ldots, n\}$, so the probability that one draw is not observation j is $1 - 1/n$. The draws are independent, so the probability that all n miss j is the product $(1 - 1/n)^n$.
Step 2: the limit. For $n = 1$, $q_1 = 0 < q_2 = \frac{1}{4}$; let $n \geqslant 2$. For $|u| < 1$, $\ln(1 - u) =$

$-\sum_{m\geqslant 1} u^m/m$, so with $u = 1/n$,

$$\ln q_n = n\ln\Big(1-\frac{1}{n}\Big) = -n\sum_{m\geqslant 1}\frac{1}{m\,n^m} = -\sum_{m\geqslant 1}\frac{1}{m\,n^{m-1}} = -1-\frac{1}{2n}-\frac{1}{3n^2}-\cdots.$$

Every term after -1 is negative and increases towards 0 as n grows, so $\ln q_n$ increases in n; and the terms after -1 sum in absolute value to at most $\sum_{m\geqslant 2} n^{-(m-1)} = 1/(n-1) \to 0$, so $\ln q_n$ tends to -1. Hence q_n increases to e^{-1}. □

Remark (ISLR §5.2)**.** Roughly a third of the observations are left out of each bootstrap sample; they are the **out-of-bag** observations on which bagging's free error estimate rests. The bootstrap and cross-validation estimate different things: cross-validation estimates a **test error**, the bootstrap the **variability of a statistic**, and it is used when no standard-error formula exists or when the formula's assumptions are in doubt.

4 Computation in R

ISLR §2.3, §5.3.1–5.3.3; §5.3.4 (the bootstrap lab), beyond the syllabus.

4.1 Data in R

The `Auto` data of `ISLR2`: nine variables on 392 cars; the package has already removed the five rows of the original file with missing values, and `na.omit` is kept for safety.

```
library(ISLR2)
dim(Auto); names(Auto)
```

```
## [1] 392   9
```

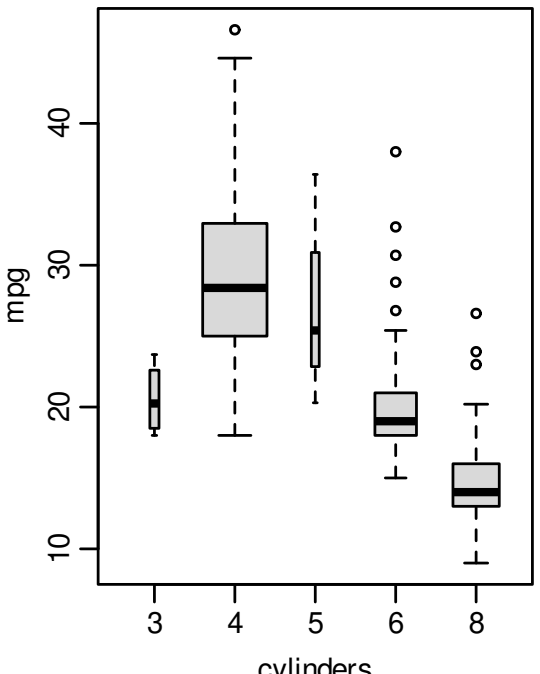
```
## [1] "mpg"          "cylinders"    "displacement"
## [4] "horsepower"   "weight"       "acceleration"
## [7] "year"         "origin"       "name"
```

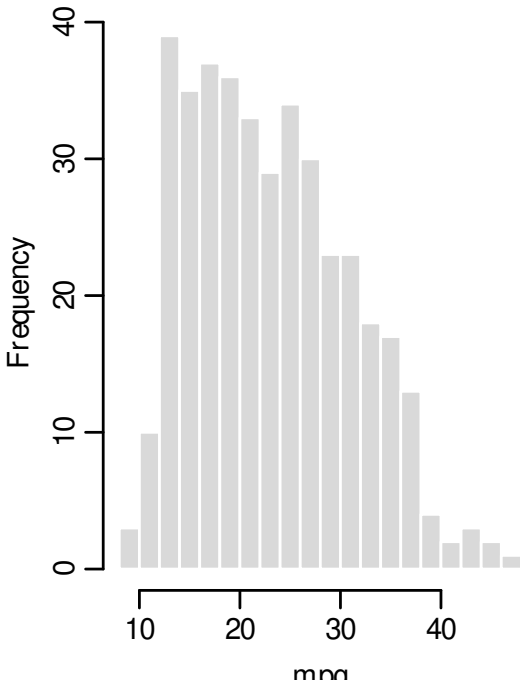
```
Auto <- na.omit(Auto)
summary(Auto[, c("mpg", "cylinders", "horsepower", "weight", "origin")])
```

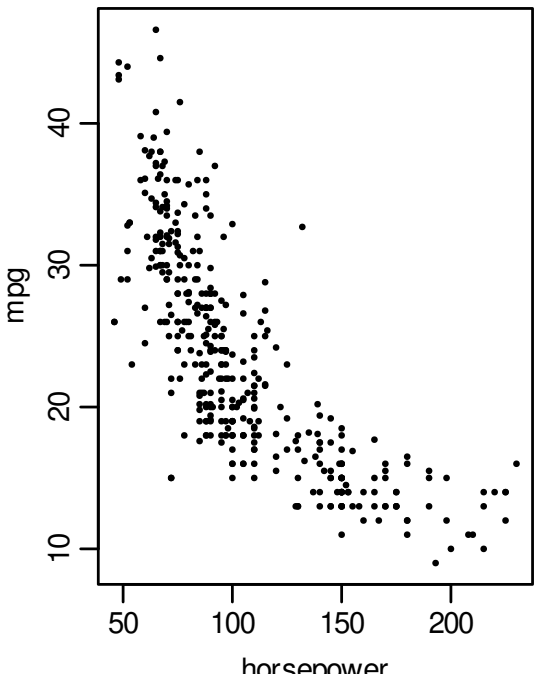
```
##       mpg          cylinders      horsepower        weight
##  Min.   : 9.0   Min.   :3.00   Min.   : 46.0   Min.   :1613
```

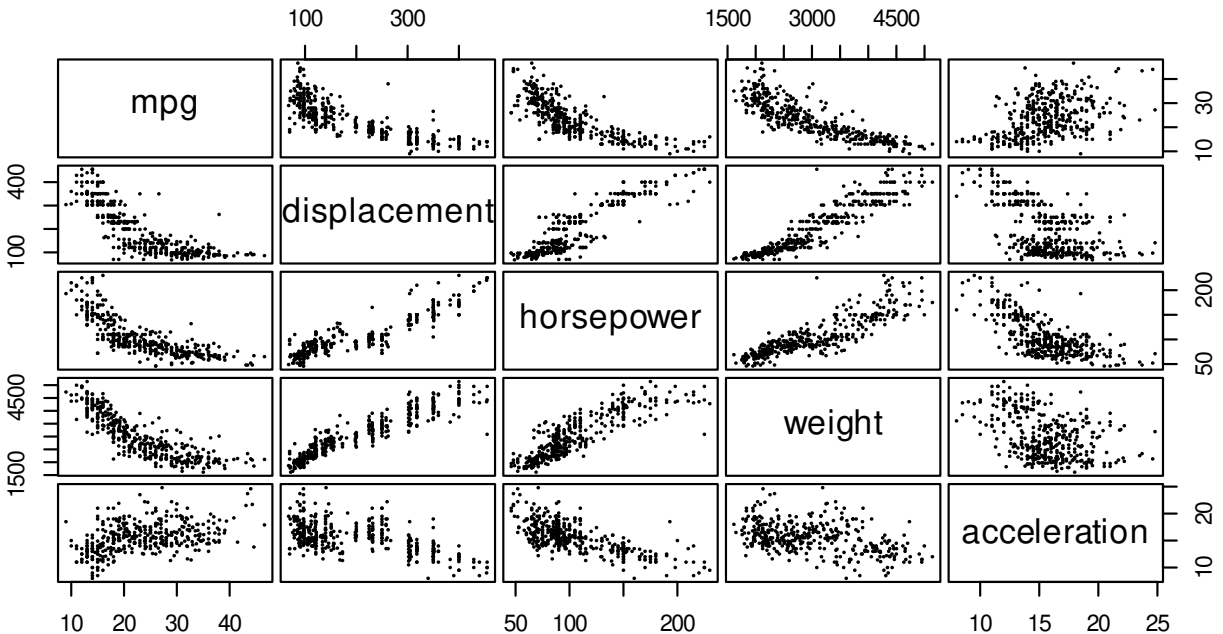
```
##  1st Qu.:17.0   1st Qu.:4.00   1st Qu.: 75.0   1st Qu.:2225
##  Median :22.8   Median :4.00   Median : 93.5   Median :2804
##  Mean   :23.4   Mean   :5.47   Mean   :104.5   Mean   :2978
##  3rd Qu.:29.0   3rd Qu.:8.00   3rd Qu.:126.0   3rd Qu.:3615
##  Max.   :46.6   Max.   :8.00   Max.   :230.0   Max.   :5140
##      origin
##  Min.   :1.00
##  1st Qu.:1.00
##  Median :1.00
##  Mean   :1.58
##  3rd Qu.:2.00
##  Max.   :3.00
```

`cylinders` is stored as a number but takes few values, so `as.factor(Auto$cylinders)` makes it qualitative and `plot` then draws box plots instead of a scatterplot; `hist` and `pairs` give a histogram and a scatterplot matrix.


mpg
10
20
30
40
3
4
5
6
8
cylinders


Frequency
0
10
20
30
40
10
20
30
40
mpg


mpg
10
20
30
40
50
100
150
200
horsepower



4.2 The validation set approach

The 392 observations are split into two halves; `mpg` is regressed on polynomials in `horsepower` on the training half and the MSE is computed on the other.

```r
set.seed(1)
train <- sample(392, 196)
val_mse <- function(deg, train)
  mean((Auto$mpg - predict(lm(mpg ~ poly(horsepower, deg), data = Auto,
                              subset = train), Auto))[-train]^2)
c(linear = val_mse(1, train), quadratic = val_mse(2, train), cubic = val_mse(3, train))
```

```
##    linear quadratic     cubic
##     23.27     18.72     18.79
```

```r
set.seed(2)
train2 <- sample(392, 196)
c(linear = val_mse(1, train2), quadratic = val_mse(2, train2), cubic = val_mse(3,
 ↪  train2))
```

```
##    linear quadratic     cubic
##     25.73     20.43     20.39
```

Both splits put the quadratic fit well below the linear one and show no gain from the cubic, and the two splits give different numbers for every model, which is drawback (1) of §3.1.

4.3 Leave-one-out cross-validation

`glm` without a `family` argument fits by least squares, exactly as `lm`, and `cv.glm` from `boot` cross-validates it.

```
library(boot); glm.fit <- glm(mpg ~ horsepower, data = Auto)
c(glm = coef(glm.fit), lm = coef(lm(mpg ~ horsepower, data = Auto)))
```

```
## glm.(Intercept)  glm.horsepower  lm.(Intercept)
##         39.9359         -0.1578         39.9359
##   lm.horsepower
##         -0.1578
```

```
cv.glm(Auto, glm.fit)$delta                    # LOOCV: both components agree
```

```
## [1] 24.23 24.23
```

```
cv.error <- sapply(1:10, function(i)
  cv.glm(Auto, glm(mpg ~ poly(horsepower, i), data = Auto))$delta[1])
round(cv.error, 3)
```

```
##  [1] 24.23 19.25 19.34 19.42 19.03 18.98 18.83 18.96 19.07
## [10] 19.49
```

The sharp drop from the linear to the quadratic fit and the absence of any clear improvement beyond it are the same conclusion as the validation set approach reached, now with no random split.

4.4 k-fold cross-validation

```
set.seed(17)
cv.error.10 <- sapply(1:10, function(i)
  cv.glm(Auto, glm(mpg ~ poly(horsepower, i), data = Auto), K = 10)$delta[1])
round(cv.error.10, 3)
```

```
##  [1] 24.27 19.27 19.35 19.30 19.03 18.90 19.12 19.15 18.87
## [10] 20.95
```

```
cv.glm(Auto, glm(mpg ~ poly(horsepower, 2), data = Auto), K = 10)$delta
```

```
## [1] 19.32 19.30
```

With `K = 10` folds the two components of **`delta`** differ slightly: the first is the fold-size-weighted average $\sum_j (n_j/n)\mathrm{MSE}_j$, with n_j the size of fold j, which is (15) when the folds have equal size (ISLR calls it the standard estimate); the second a bias-corrected version.

4.5 The shortcut formula

Equation (12) from one fit, against the n explicit refits of `cv.glm`. `cv.glm` does not use the shortcut, so the agreement is a check of the formula rather than of the software.

```
fit <- lm(mpg ~ poly(horsepower, 2), data = Auto)
h <- hatvalues(fit); e <- residuals(fit)
c(shortcut = mean((e / (1 - h))^2), cv.glm = cv.error[2],
  training_MSE = mean(e^2), min_h = min(h), max_h = max(h))
```

```
##     shortcut       cv.glm training_MSE        min_h
##    19.248213    19.248213    18.984769     0.003231
##        max_h
##     0.093597
```

4.6 The bootstrap (beyond the syllabus)

`boot(data, statistic, R)` draws `R` bootstrap samples and applies `statistic(data, index)` to each; the function must take the row indices as its second argument. Here the statistic is the pair of least squares coefficients of `mpg` on `horsepower`.

```
boot.fn <- function(data, index)
  coef(lm(mpg ~ horsepower, data = data, subset = index))
```

```r
boot.fn(Auto, 1:392)
```

```
## (Intercept)  horsepower
##     39.9359     -0.1578
```

```r
set.seed(1)
bo <- boot(Auto, boot.fn, R = 1000)
rbind(bootstrap_SE = apply(bo$t, 2, sd),
      formula_SE   = summary(lm(mpg ~ horsepower, data = Auto))$coef[, "Std. Error"])
```

```
##              (Intercept) horsepower
## bootstrap_SE      0.8439   0.007367
## formula_SE        0.7175   0.006446
```

The formula standard errors of Unit 2 assume the linear model is correct with constant error variance; the bootstrap assumes neither, and the two disagree here because the true relationship is not linear, so the residual variance varies with **horsepower**. Bootstrapping the quadratic fit, which is closer to the truth, brings them closer:

```r
boot.fn2 <- function(data, index)
  coef(lm(mpg ~ horsepower + I(horsepower^2), data = data, subset = index))
set.seed(1)
```

```r
bo2 <- boot(Auto, boot.fn2, R = 1000)
rbind(bootstrap_SE = apply(bo2$t, 2, sd),
      formula_SE = summary(lm(mpg ~ horsepower + I(horsepower^2), data = Auto))$coef[,
       ↪  2])
```

```
##                (Intercept) horsepower I(horsepower^2)
## bootstrap_SE          2.03    0.03242        0.0001172
## formula_SE            1.80    0.03112        0.0001221
```

5 Exercises

Exercise 1.1 (ISLR §2.4, exercise 1)**.** For each scenario, say whether a flexible method is expected to perform better or worse than an inflexible one, and why. (a) n extremely large and p small. (b) p extremely large and n small. (c) The relationship between predictors and response is highly non-linear. (d) $\text{var}(\varepsilon)$ is extremely high.

Solution. Read each through (7). (a) Better. A flexible method lowers the bias, and with n large its variance is small, so the sum falls. (b) Worse. With few observations and many predictors the flexible method's variance is large; it fits the noise and overfits. (c) Better. An inflexible method

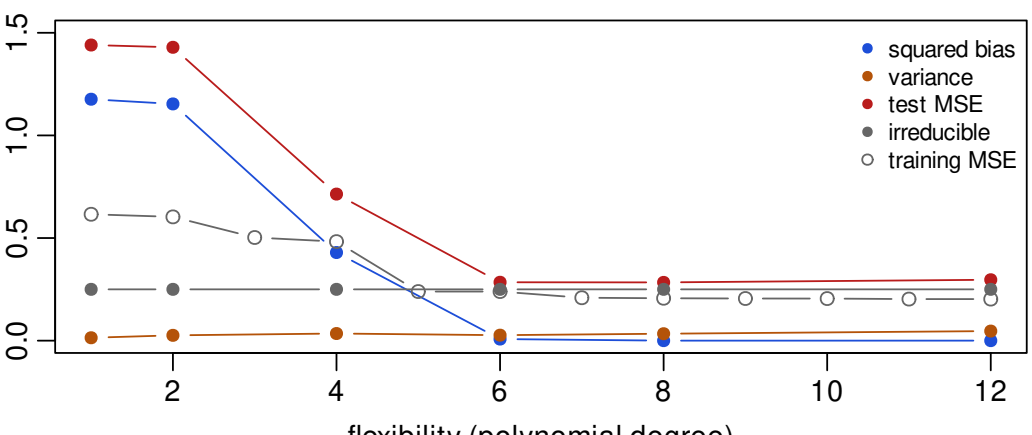
cannot represent the non-linearity and carries a large squared bias that no sample size removes. (d) Worse. The noise dominates the signal, and a flexible method follows the noise; its variance is high while the bias reduction it buys is worth little. ■

Exercise 1.2 (ISLR §2.4, exercise 2)**.** For each scenario, say whether it is a classification or a regression problem, whether the interest is inference or prediction, and give n and p. (a) Data on the top 500 US firms: profit, number of employees, industry and CEO salary; the aim is to understand which factors affect CEO salary. (b) Twenty similar products previously launched, each recorded as a success or failure together with price, marketing budget, competition price and ten other variables; the aim is to predict whether a new product will succeed. (c) Weekly data for 2012 on the % change in the USD/Euro exchange rate and the % changes in the US, British and German markets; the aim is to predict the exchange rate change.

Solution. (a) Regression, since salary is quantitative; inference, since the question is which factors affect it; $n = 500$, $p = 3$ (profit, employees, industry). (b) Classification, the response being success or failure; prediction; $n = 20$, $p = 13$. (c) Regression, the response being a percentage change; prediction; $n = 52$ weeks, $p = 3$. ■

Exercise 1.3 (ISLR §2.4, exercise 3)**.** Sketch, on one plot against flexibility, the typical squared bias, variance, training error, test error and Bayes (irreducible) error curves, and explain the shape of each.

Solution. The five curves, drawn as functions of the polynomial degree: the squared bias, variance, test MSE and irreducible error at $x_0 = 1$ come from the simulation in §2.2 (training sets of 60 observations), and the training MSE, averaged over the training points, from the simulation in §2.1 (80 observations):



Squared bias decreases: a more flexible method can approximate f more closely. Variance increases: a more flexible fit changes more from one training set to another. Training MSE decreases monotonically, since flexibility lets the fit follow the training points. Test MSE is the sum of squared bias, variance and the irreducible error by (7), so it falls while the bias falls faster than the variance rises and then rises: a U. The irreducible error $\operatorname{var}(\varepsilon)$ is a horizontal line, the floor under the test MSE. ■

Exercise 1.4 (ISLR §2.4, exercise 4)**.** Describe three real-life applications each of classification, regression and cluster analysis, giving the response and the predictors and saying whether the goal is inference or prediction.

Solution. Classification. (i) Claim fraud: response fraudulent or not; predictors claim size, delay in reporting, provider, prior claims; prediction, to route claims for investigation. (ii) Policy lapse: response lapse in the next year or not; predictors age, premium change, tenure, payment method; both, since the insurer wants to predict lapses and to know which factors drive them. (iii) Credit default: response default or not; predictors balance, income, student status; prediction.
Regression. (i) Claim severity: response amount paid; predictors driver age, vehicle use, region;

inference, since the rating factors are the object (FREES §1.6). (ii) House value: response price; predictors size, location, crime rate; prediction of value, or inference about the river-view premium. (iii) Reserving: response ultimate loss; predictors accident year, development year, exposure; prediction.

Clustering. (i) Market segmentation of policyholders by income, age, coverage and claims history, with no response, to identify groups for product design. (ii) Grouping territories by loss experience and demographics before rating. (iii) Grouping firms by financial ratios to find distressed profiles. ■

Exercise 1.5 (ISLR §2.4, exercise 5)**.** What are the advantages and disadvantages of a very flexible approach for regression or classification, relative to a less flexible one? When is each preferred?

Solution. A flexible method can represent a wide range of shapes, so it has low bias and can fit a non-linear f that an inflexible method cannot; its disadvantages are high variance, the need to choose a tuning level such as smoothness or K, a larger sample requirement, and low interpretability. By (7) it is preferred when the truth is far from the inflexible form and n is large

enough for the variance to be small, and when the goal is prediction alone. An inflexible method is preferred when the goal is inference, when n is small or p large relative to n, when the noise is large, or when the truth is close to the assumed form, in which case the flexible method pays variance for no bias reduction. ■

Exercise 1.6 (ISLR §2.4, exercise 6)**.** Describe the difference between a parametric and a non-parametric approach, and the advantages and disadvantages of the parametric one.

Solution. A parametric approach assumes a functional form for f and reduces estimation to a finite set of parameters; a non-parametric approach assumes no form and estimates f directly subject to a smoothness constraint (Definitions 1.5 and 1.6). The parametric approach needs fewer observations, is easy to fit and to interpret, and yields the inferential machinery of Units 2 and 3; its disadvantage is that the assumed form may be far from the true f, in which case the estimate is biased however large the sample. The non-parametric approach removes that risk at the cost of far more data, a smoothness choice that can overfit, and a fit that is hard to summarise. ■

Exercise 1.7 (ISLR §2.4, exercise 7)**.** The training set below has three predictors and a qualitative response.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	−1	0	1	Green
6	1	1	1	Red

(a) Compute the Euclidean distance from each observation to the test point $X_1 = X_2 = X_3 = 0$. (b) Give the prediction with $K = 1$ and with $K = 3$, with reasons. (c) If the Bayes decision boundary is highly non-linear, should the best K be large or small?

Solution. (a) The distances are $\sqrt{0+9+0} = 3$, $\sqrt{4} = 2$, $\sqrt{1+9} = \sqrt{10} = 3.162$, $\sqrt{1+4} = \sqrt{5} = 2.236$, $\sqrt{1+0+1} = \sqrt{2} = 1.414$ and $\sqrt{3} = 1.732$.

```r
Xk <- rbind(c(0,3,0), c(2,0,0), c(0,1,3), c(0,1,2), c(-1,0,1), c(1,1,1))
Yk <- c("Red","Red","Red","Green","Green","Red")
dk <- sqrt(rowSums(Xk^2))
cbind(obs = 1:6, dist = round(dk, 4), Y = Yk)[order(dk), ]
```

```
##      obs dist     Y
## [1,] "5" "1.4142" "Green"
## [2,] "6" "1.7321" "Red"
## [3,] "2" "2"      "Red"
## [4,] "4" "2.2361" "Green"
## [5,] "1" "3"      "Red"
## [6,] "3" "3.1623" "Red"
```

(b) $K = 1$: the nearest observation is 5, Green, so predict Green. $K = 3$: the three nearest are 5 (Green), 6 (Red) and 2 (Red); by (10) the estimated probability of Red is 2/3, so predict Red.
(c) Small. By Definition 1.19 the rule averages over the K nearest points, so its boundary can bend only on the scale of that neighbourhood; a highly non-linear Bayes boundary needs small neighbourhoods, hence small K, at the price of higher variance. ■

Exercise 1.8 (ISLR §2.4, exercise 8)**.** Using the `College` data (777 US colleges, 18 variables): (i) produce a numerical summary; (ii) a scatterplot matrix of the first ten columns; (iii) side-by-side box plots of `Outstate` against `Private`; (iv) create `Elite` by binning `Top10perc` at 50%, count the elite colleges and box-plot `Outstate` against `Elite`; (v) histograms of some quantitative variables with differing numbers of bins; (vi) summarise what you find.

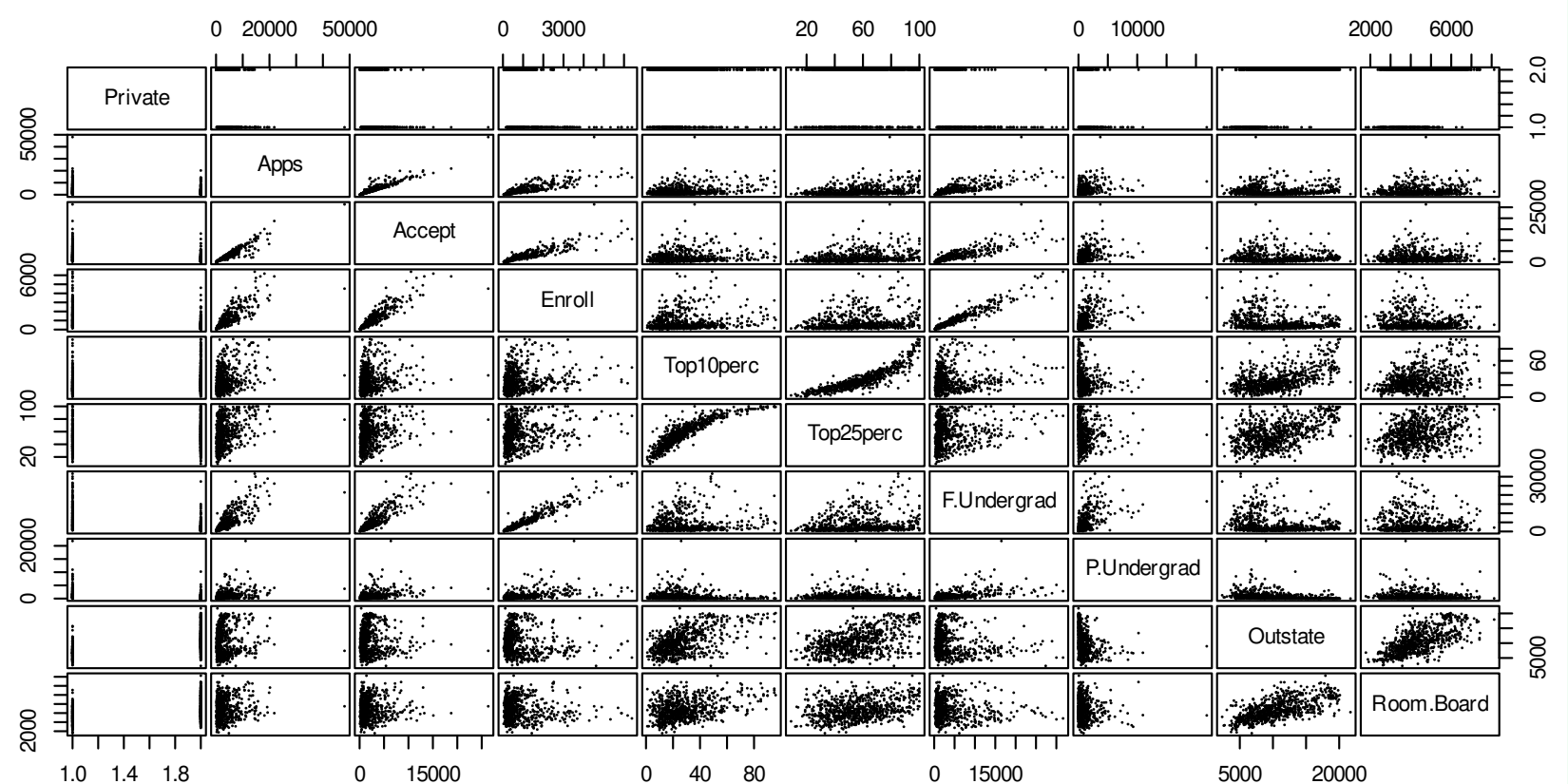
Solution. In `ISLR2` the college names are already row names, so the `read.csv` and `rownames` steps of the printed exercise are not needed. The numerical summary below is abridged: it covers the first six of the 18 columns, and its printed output is truncated; `summary(college)` gives all of them.

```
college <- ISLR2::College
dim(college)
```

```
## [1] 777  18
```

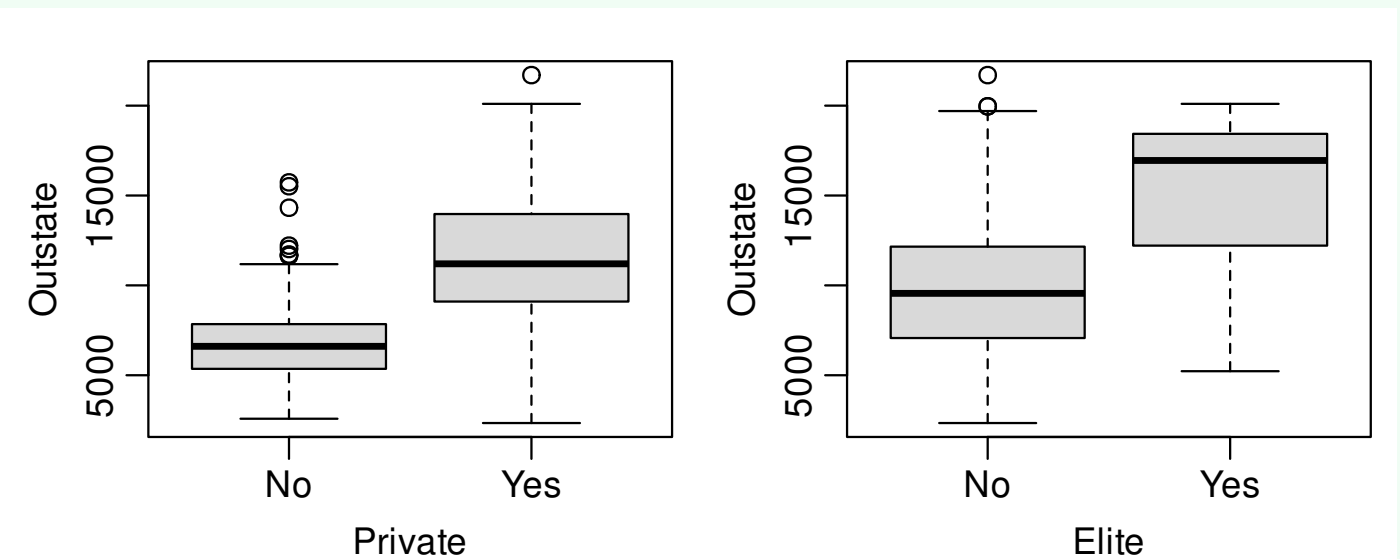
```r
summary(college[, 1:6])
```

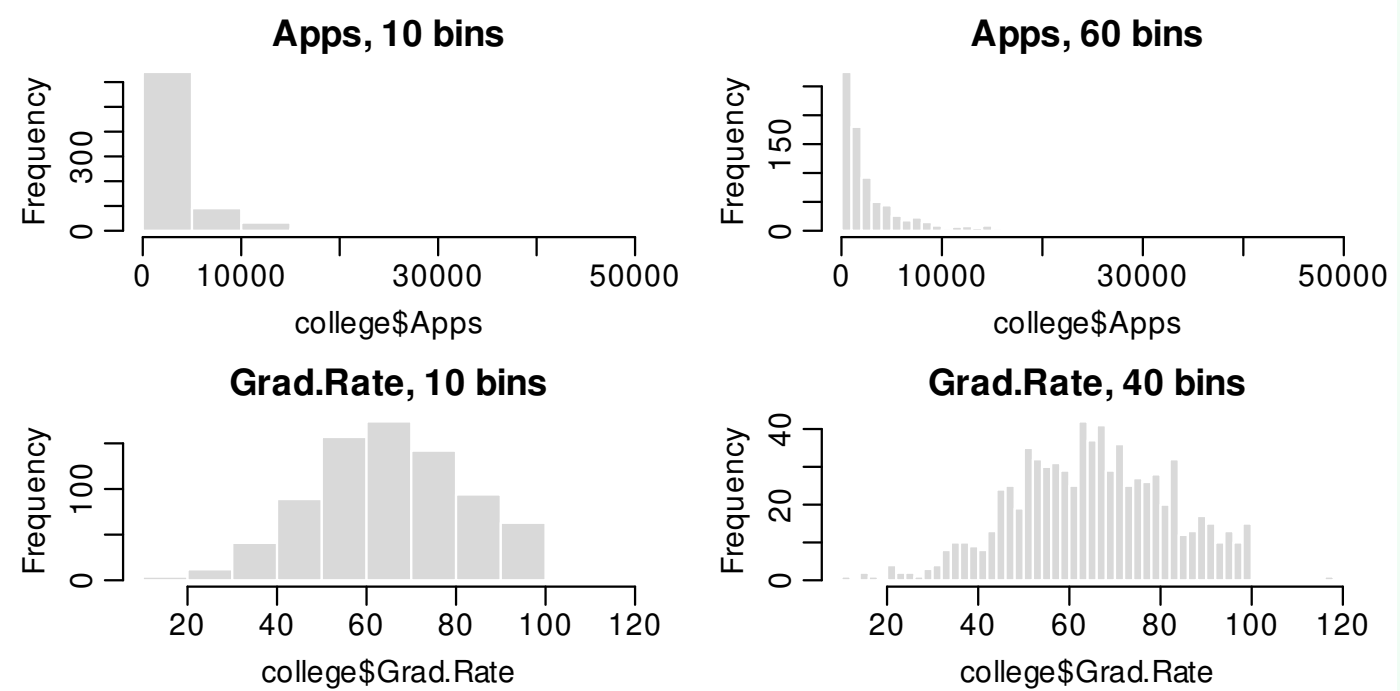
```
##  Private        Apps           Accept          Enroll
##  No :212   Min.   :   81   Min.   :   72   Min.   :  35
##  Yes:565   1st Qu.:  776   1st Qu.:  604   1st Qu.: 242
##            Median : 1558   Median : 1110   Median : 434
##            Mean   : 3002   Mean   : 2019   Mean   : 780
##            3rd Qu.: 3624   3rd Qu.: 2424   3rd Qu.: 902
##            Max.   :48094   Max.   :26330   Max.   :6392
##    Top10perc       Top25perc
##  Min.   : 1.0   Min.   :  9.0
##  1st Qu.:15.0   1st Qu.: 41.0
##  Median :23.0   Median : 54.0
##  Mean   :27.6   Mean   : 55.8
## ... [output truncated]
```


Private
Apps
Accept
Enroll
Top10perc
Top25perc
F.Undergrad
P.Undergrad
Outstate
Room.Board

```r
college$Elite <- factor(ifelse(college$Top10perc > 50, "Yes", "No"))
table(college$Elite)
```

```
##
##  No Yes
## 699  78
```






```
c(private_share = mean(college$Private == "Yes"),
  grad_rate_over_100 = sum(college$Grad.Rate > 100),
  cor_outstate_top10 = cor(college$Outstate, college$Top10perc),
  cor_accept_rate_outstate = cor(college$Accept/college$Apps, college$Outstate))
```

```
##           private_share       grad_rate_over_100
##                  0.7272                   1.0000
##      cor_outstate_top10 cor_accept_rate_outstate
##                  0.5623                  -0.2410
```

(vi) Private colleges charge more out-of-state tuition than public ones and the elite colleges more still, with the box plots barely overlapping. `Apps` is heavily right-skewed, a few universities receiving tens of thousands of applications, and the finer histogram shows that; the coarse one hides it. One graduation rate exceeds 100%, a data error to be treated before any modelling. Out-of-state tuition rises with the share of top-decile students and falls with the acceptance rate, so selectivity and price move together. ■

Exercise 1.9 (ISLR §2.4, exercise 9)**.** For the `Auto` data with missing values removed: (a) which predictors are quantitative and which qualitative; (b) the range of each quantitative predictor; (c) the mean and standard deviation of each; (d) the same after removing observations 10 to 85; (e) investigate the predictors graphically; (f) which variables might predict `mpg`?

Solution. (a) **mpg**, **displacement**, **horsepower**, **weight**, **acceleration** and **year** are quantitative; **origin** is qualitative though coded 1, 2, 3 (American, European, Japanese), **cylinders** takes five values and is best treated as qualitative, and **name** is a label.

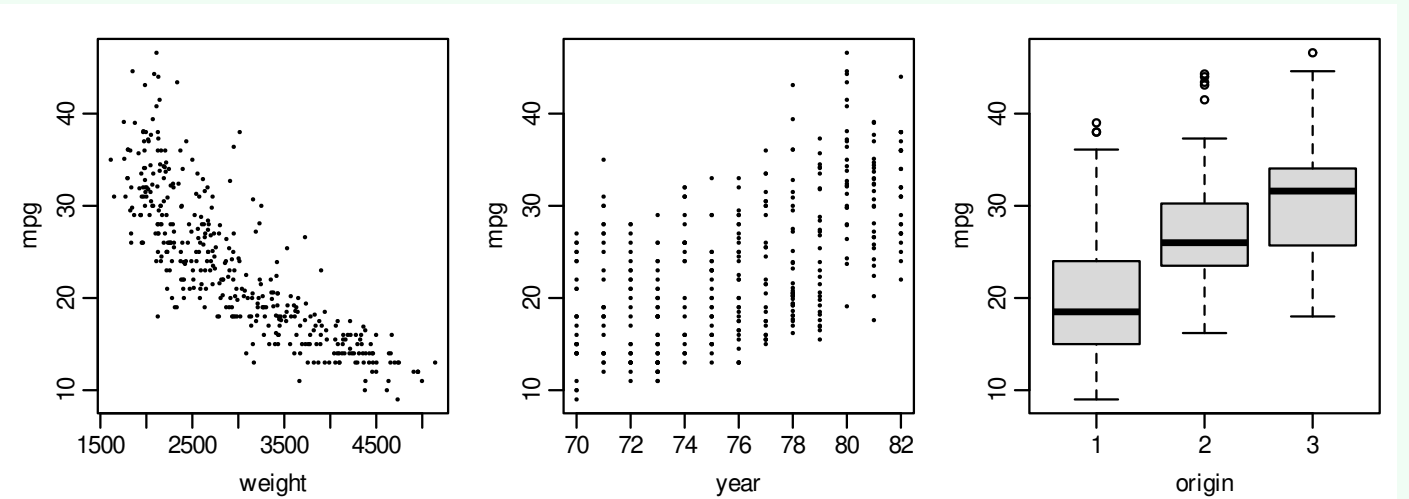
```
quant <- c("mpg", "displacement", "horsepower", "weight", "acceleration", "year")
stats <- function(d) rbind(min = sapply(d, min), max = sapply(d, max),
                           mean = round(sapply(d, mean), 2), sd = round(sapply(d,
                            ↪  sd), 2))
stats(Auto[, quant])                                          # (b), (c)
```

```
##        mpg displacement horsepower weight acceleration  year
## min   9.00         68.0      46.00 1613.0         8.00 70.00
## max  46.60        455.0     230.00 5140.0        24.80 82.00
## mean 23.45        194.4     104.47 2977.6        15.54 75.98
## sd    7.81        104.6      38.49  849.4         2.76  3.68
```

```
stats(Auto[-(10:85), quant])                                  # (d)
```

```
##        mpg displacement horsepower weight acceleration  year
## min  11.00        68.00      46.00 1649.0         8.50 70.00
## max  46.60       455.00     230.00 4997.0        24.80 82.00
```

```
## mean 24.40       187.24     100.72 2936.0       15.73 77.15
## sd    7.87        99.68      35.71  811.3        2.69  3.11
```




```
round(cor(Auto[, quant])["mpg", ], 3)
```

```
##          mpg displacement   horsepower       weight
##        1.000       -0.805       -0.778       -0.832
## acceleration         year
##        0.423        0.581
```

(e), (f) Fuel economy falls steeply and non-linearly with weight, displacement and horsepower,

which are strongly correlated with one another; it rises with model year, by about a mile per gallon a year over the period; and Japanese and European cars have higher `mpg` than American ones, largely because they are lighter. Weight, or one of its correlates, together with year and origin would predict `mpg` well; the curvature in the weight plot says a transformation or a non-linear method is needed for the best fit. ■

Exercise 1.10 (ISLR §2.4, exercise 10)**.** For the `Boston` housing data: (a) how many rows and columns, and what do they represent; (b) pairwise scatterplots and what they show; (c) which predictors are associated with per-capita crime rate; (d) do some suburbs have particularly high crime, tax rates or pupil–teacher ratios; (e) how many suburbs bound the Charles river; (f) the median pupil–teacher ratio; (g) the suburb with the lowest median home value, and how its other predictors compare with the overall ranges; (h) how many suburbs average more than seven rooms per dwelling, more than eight, and what characterises the latter.

Solution.

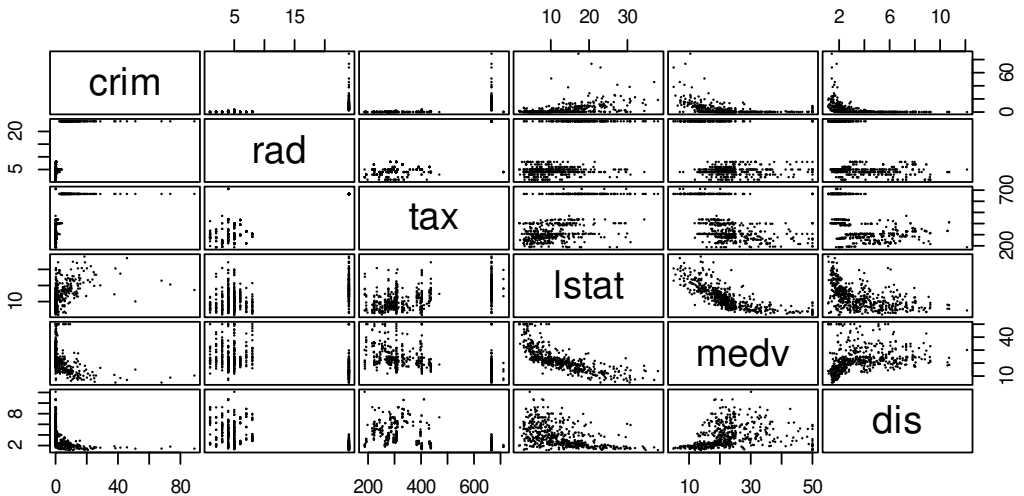
```
B <- ISLR2::Boston
dim(B)                                                    # (a)
```

```
## [1] 506  13
```

```
round(sort(cor(B)["crim", -1], decreasing = TRUE), 3)      # (c)
```

```
##     rad     tax   lstat     nox   indus     age ptratio
##   0.626   0.583   0.456   0.421   0.407   0.353   0.290
##    chas      zn      rm     dis    medv
##  -0.056  -0.200  -0.219  -0.380  -0.388
```

(a) 506 rows, one per census tract in the Boston area, and 13 columns: the per-capita crime rate **crim**, eleven tract characteristics (zoning, industry, the Charles river indicator, nitrogen oxides, rooms, age, distance to employment centres, highway access, tax rate, pupil–teacher ratio, lower-status share) and the median home value **medv**.



(b), (c) Crime is highest in tracts with high highway accessibility, high tax rates, high lower-status share and low home values, and low in tracts far from employment centres; all the crime relationships are non-linear, most tracts having near-zero crime and a few very high values.

```
rbind(crim = quantile(B$crim, c(0, .5, .9, .99, 1)),
      tax = quantile(B$tax, c(0, .5, .9, .99, 1)),
```

```
      ptratio = quantile(B$ptratio, c(0, .5, .9, .99, 1)))      # (d)
```

```
##                0%      50%    90%    99%   100%
## crim      0.00632   0.2565  10.75  41.37  88.98
## tax     187.00000 330.0000 666.00 666.00 711.00
## ptratio  12.60000  19.0500  20.90  21.20  22.00
```

```
c(chas = sum(B$chas == 1), median_ptratio = median(B$ptratio))  # (e), (f)
```

```
##           chas median_ptratio
##          35.00          19.05
```

(d) Crime is extremely right-skewed: the median tract has a rate near zero and the upper tail runs to several times the ninetieth percentile. Tax rates and pupil–teacher ratios show a cluster of tracts at the top of the range rather than isolated extremes. (e), (f) The counts print above.

```
low <- which.min(B$medv)
rbind(lowest = B[low, ], min = sapply(B, min), median = sapply(B, median), max =
 ↪  sapply(B, max))
```

```
##            crim  zn indus chas   nox    rm   age    dis rad
## lowest 38.35180   0 18.10    0 0.693 5.453 100.0  1.490  24
## min     0.00632   0  0.46    0 0.385 3.561   2.9  1.130   1
```

```
## median  0.25651   0  9.69    0 0.538 6.208  77.5  3.207   5
## max    88.97620 100 27.74    1 0.871 8.780 100.0 12.127  24
##        tax ptratio lstat medv
## lowest 666   20.20 30.59  5.0
## min    187   12.60  1.73  5.0
## median 330   19.05 11.36 21.2
## max    711   22.00 37.97 50.0
```

(g) Two tracts, rows 399 and 406, tie at the minimum **medv**= 5, and **which.min** reports only the first; row 406 has the same **age**, **rad**, **tax** and **ptratio**, a higher crime rate (67.9), a lower lower-status share (23.0) and slightly more rooms (5.68). The lowest-value tract printed has the maximum **age** and **rad**, tax and pupil–teacher ratios at the top of their ranges, a high crime rate and lower-status share, few rooms and is close to the employment centres: every predictor points the same way.

```
c(rooms_over_7 = sum(B$rm > 7),
  rooms_over_8 = sum(B$rm > 8))
```

```
## rooms_over_7 rooms_over_8
##           64           13
```

```
round(rbind(over_8 = colMeans(B[B$rm > 8, ]), all = colMeans(B)), 2)
```

```
##         crim    zn indus chas  nox   rm   age  dis  rad   tax
## over_8 0.72 13.62  7.08 0.15 0.54 8.35 71.54 3.43 7.46 325.1
## all    3.61 11.36 11.14 0.07 0.55 6.28 68.57 3.80 9.55 408.2
##         ptratio lstat  medv
## over_8    16.36  4.31 44.20
## all       18.46 12.65 22.53
```

(h) The tracts averaging more than eight rooms have far higher median home values, lower crime and lower-status shares, and lower pupil–teacher ratios than the average tract. ■

Exercise 1.11 (FREES exercise 1.1)**.** The file `HealthExpend` holds 2,000 individuals aged 18 to 65 from panels 7 and 8 of the 2003 Medical Expenditure Panel Survey, with their inpatient (`EXPENDIP`) and outpatient (`EXPENDOP`) expenditures. Part 1, on the 157 individuals with positive inpatient expenditure: (a) descriptive statistics of `EXPENDIP`: the typical (mean and median) expenditure, the standard deviation against the mean, and whether the data appear skewed; (b) a box plot, a histogram and a normal *qq* plot; (c) the square-root and the logarithmic transform, each summarised by a histogram and a *qq* plot: is either approximately normal? Part 2, on the

1,352 individuals with positive outpatient expenditure: (d) repeat (a), and compare histograms of the expenditures and of their logarithms.

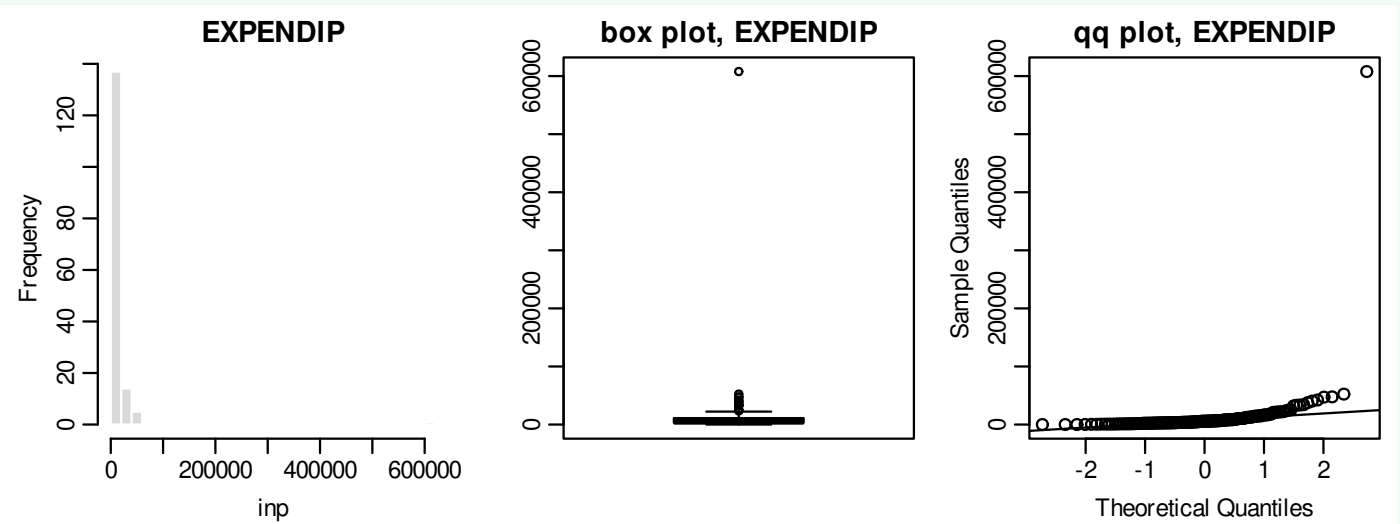
Solution.

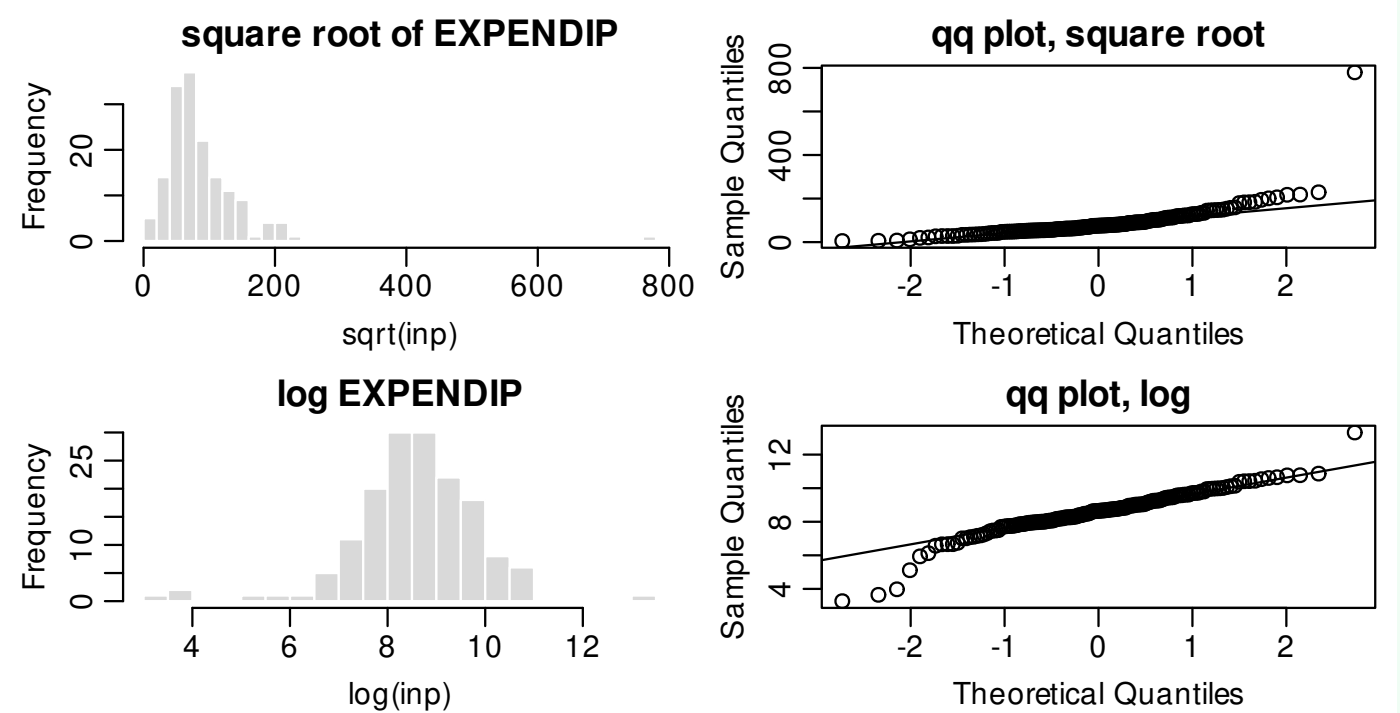
```
meps <- read.csv("data/frees/csv/HealthExpend.csv")
inp  <- meps$EXPENDIP[meps$EXPENDIP > 0]
outp <- meps$EXPENDOP[meps$EXPENDOP > 0]
descr <- function(v) c(n = length(v), mean = mean(v), median = median(v), sd =
 ↪  sd(v),
                       min = min(v), max = max(v), skew = skew(v))
round(rbind(inpatient = descr(inp), outpatient = descr(outp)), 2)
```

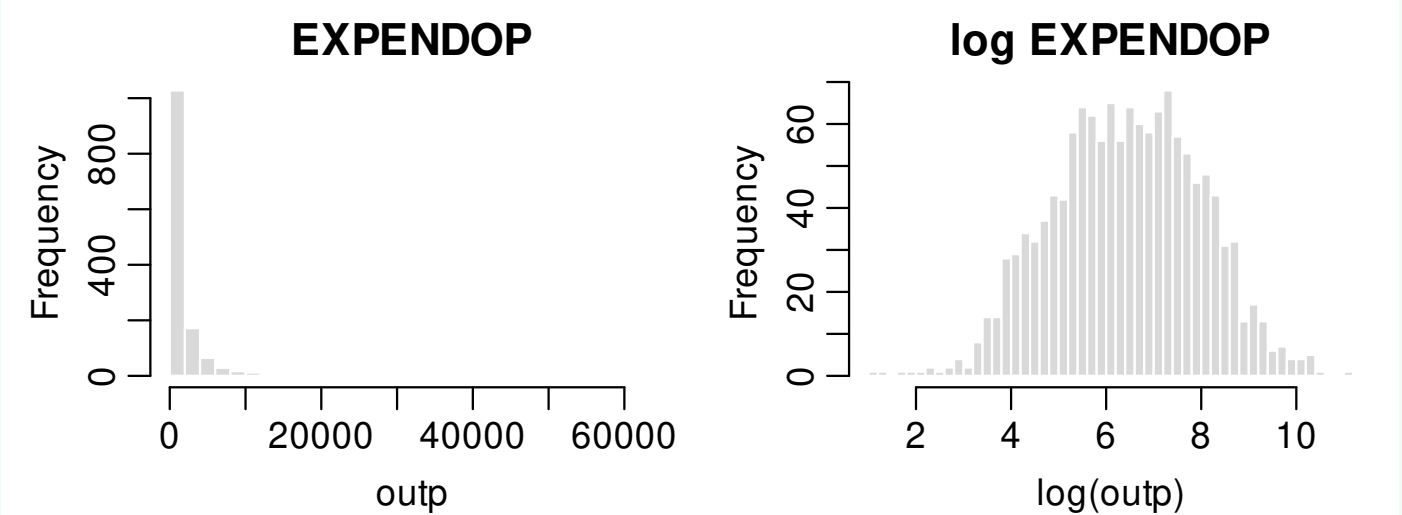
```
##                n  mean median    sd   min    max  skew
## inpatient    157 12844 5694.7 48837 26.31 607801 11.64
## outpatient  1352  1854  625.7  3864  3.00  62811  6.50
```

```
round(rbind(inpatient  = c(raw = skew(inp),  sqrt = skew(sqrt(inp)),  log =
 ↪  skew(log(inp))),
            outpatient = c(raw = skew(outp), sqrt = skew(sqrt(outp)), log =
             ↪  skew(log(outp)))), 3)
```

```
##                raw  sqrt    log
## inpatient   11.636 6.000 -0.790
## outpatient   6.501 2.225 -0.072
```




square root of EXPENDIP
Frequency
0
20
0
200
400
600
800
sqrt(inp)
qq plot, square root
Sample Quantiles
0
400
800
-2
-1
0
1
2
Theoretical Quantiles
log EXPENDIP
Frequency
0
10
25
4
6
8
10
12
log(inp)
qq plot, log
Sample Quantiles
4
8
12
-2
-1
0
1
2
Theoretical Quantiles



(a) The mean inpatient expenditure, \$12,844, is more than twice the median, \$5,695, and the standard deviation is nearly four times the mean: the distribution is strongly skewed to the right (skewness 11.6). (b)–(c) The histogram is a spike at small amounts with a tail to \$607,801, and the *qq* plot rises far above the line on the right. The square root leaves a skewness of 6.0; the logarithm brings it to -0.8, and its *qq* plot is close to the line except in the lower tail. (d) Outpatient expenditures: mean \$1,854, median \$626, skewness 6.5; their logarithm, with skewness

−0.07, has the most nearly symmetric histogram of all six distributions. ■

Exercise 1.12 (FREES exercise 1.2)**.** The file `WiscNursingHome` holds the annual cost reports of Wisconsin Medicaid-certified nursing homes for 2000 (362 facilities) and 2001 (355). The outcome is total patient years `TPY`, patient days divided by operating days; the number of beds `NUMBED` and the square footage `SQRFOOT` measure the size of a facility. Part 1, on 2000: (a) descriptive statistics of `TPY`, `NUMBED` and `SQRFOOT`; (b) a histogram and a *qq* plot of `TPY`: approximately normal? (c) the same for ln `TPY`. Part 2: repeat (a) and (c) for 2001.

Solution.

```
nh <- read.csv("data/frees/csv/WiscNursingHome.csv")
nh2000 <- subset(nh, CRYEAR == 2000)
nh2001 <- subset(nh, CRYEAR == 2001)
descr_na <- function(v) descr(v[!is.na(v)])
round(sapply(nh2000[, c("TPY", "NUMBED", "SQRFOOT")], descr_na), 2)
```

```
##            TPY NUMBED SQRFOOT
## n       362.00 362.00  357.00
## mean     88.79  97.08   50.14
```

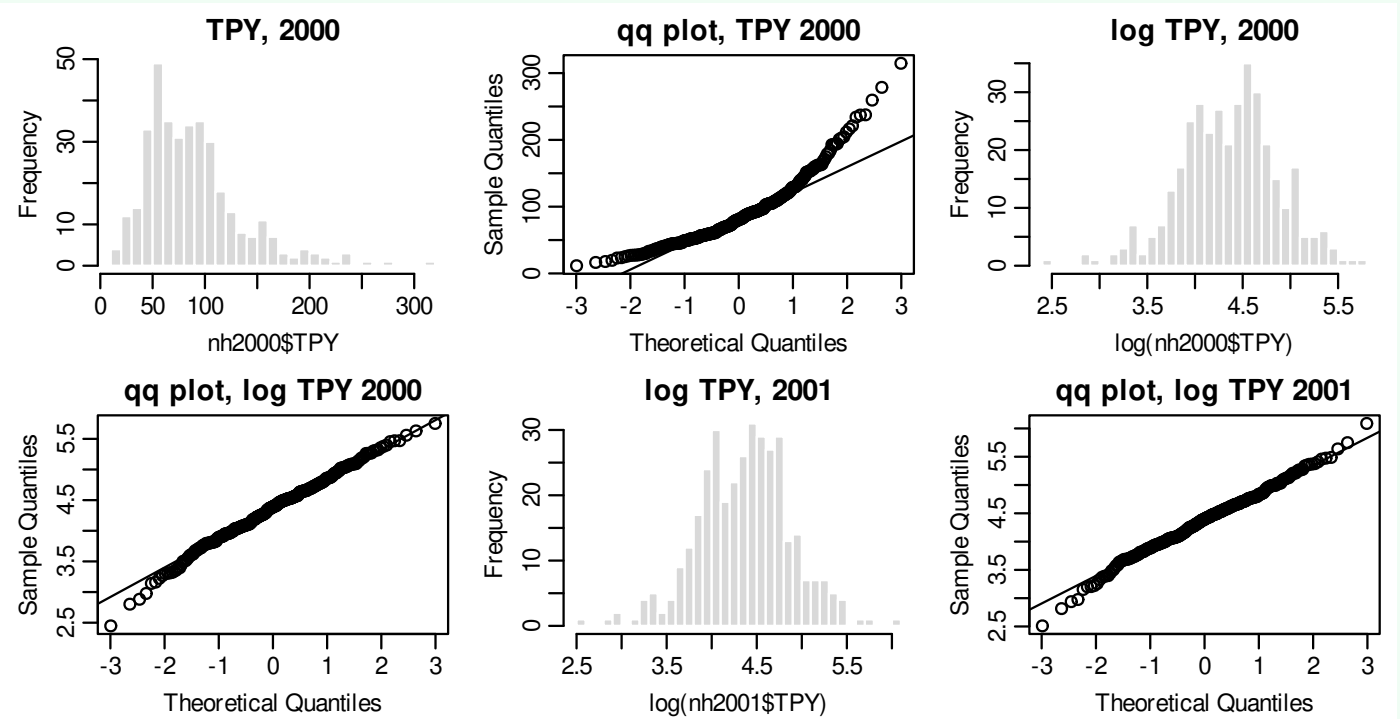
```
## median  80.54  90.00   39.22
## sd      46.10  48.99   34.50
## min     11.57  18.00    5.64
## max    314.70 320.00  262.00
## skew     1.40   1.35    2.22
```

```
round(sapply(nh2001[, c("TPY", "NUMBED", "SQRFOOT")], descr_na), 2)
```

```
##            TPY NUMBED SQRFOOT
## n       355.00 355.00  350.00
## mean     89.71  97.33   50.37
## median   81.13  90.00   40.26
## sd       49.05  51.97   35.56
## min      12.31  18.00    5.64
## max     440.67 457.00  262.00
## skew      2.10   2.08    2.38
```

```
c(log_TPY_2000 = skew(log(nh2000$TPY)), log_TPY_2001 = skew(log(nh2001$TPY)))
```

```
## log_TPY_2000 log_TPY_2001
##      -0.2417      -0.1423
```



(a) In 2000 the median facility had 80.5 patient years and 90 beds; `SQRFOOT` is missing for five facilities. All three size measures are skewed to the right (skewness 1.4, 1.35 and 2.2). (b) The histogram of `TPY` has a long right tail and its *qq* plot curves upward at the right. (c) ln `TPY` has

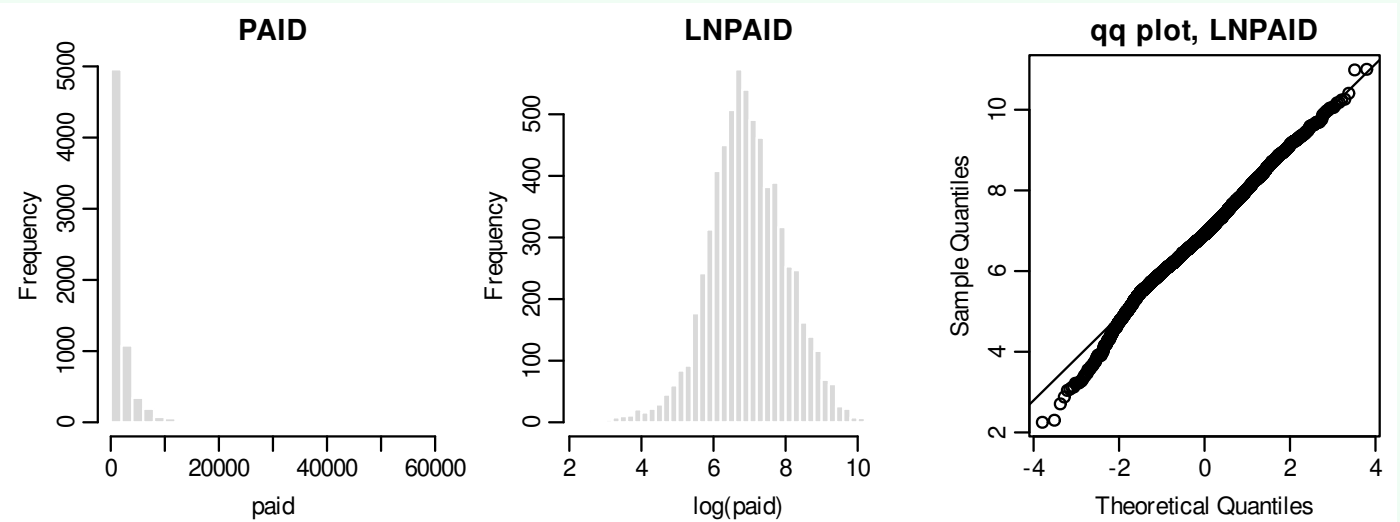
skewness -0.24 and a nearly straight qq plot: approximately normal. Part 2: 2001 shows the same pattern with a larger maximum (440.7 patient years, 457 beds), which raises the skewness of `TPY` to 2.1; its logarithm has skewness -0.14. ■

Exercise 1.13 (FREES exercise 1.3)**.** The file `AutoClaims` holds $n = 6{,}773$ private passenger automobile claims of drivers aged 50 and over, with the state code, rating class, sex, age and the amount `PAID` to settle the claim. Examine the histogram of `PAID` and comment on its symmetry; create `LNPAID` $= \ln$ `PAID` and comment on the symmetry of its histogram and qq plot.

Solution.

```
auto_claims <- read.csv("data/frees/csv/AutoClaims.csv")
paid <- auto_claims$PAID
round(rbind(PAID = descr(paid), LNPAID = descr(log(paid))), 3)
```

```
##            n     mean   median       sd   min   max   skew
## PAID    6773 1853.035 1001.700 2646.909 9.500 60000  6.236
## LNPAID 6773    6.956    6.909    1.071 2.251    11 -0.086
```



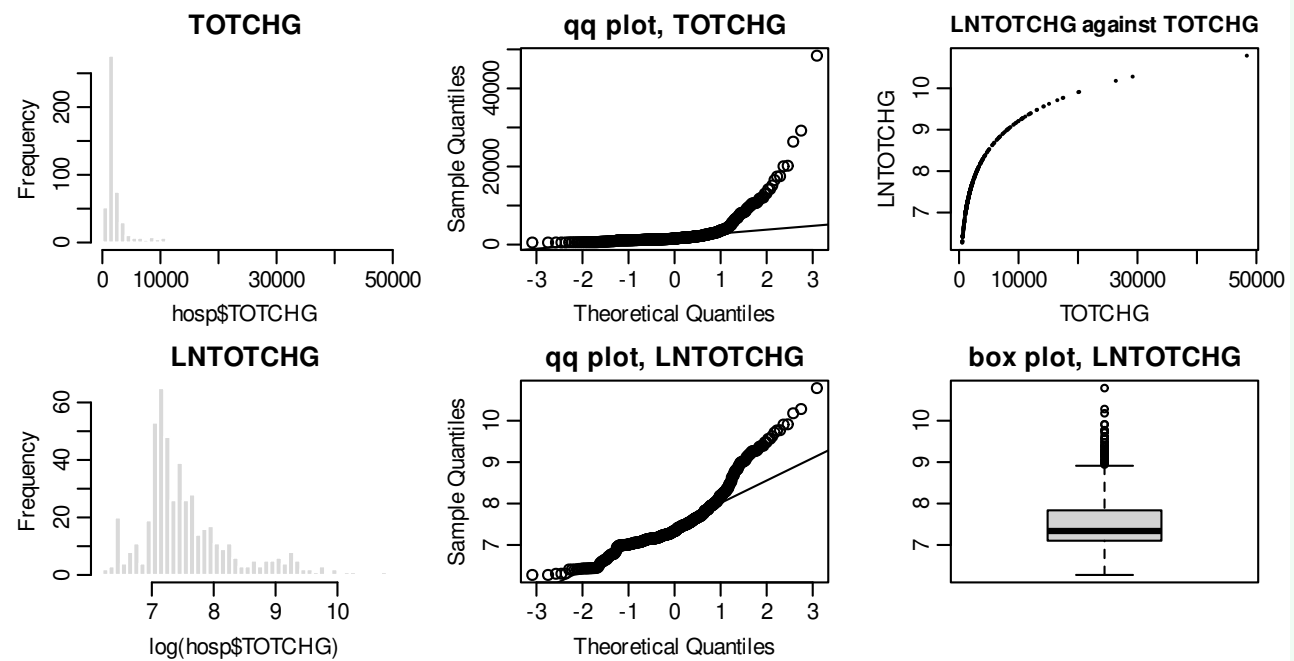
The mean amount paid, \$1,853, is nearly twice the median, \$1,002, and the largest claim is \$60,000: the histogram is a spike at small amounts with a long right tail (skewness 6.2). The logarithm is nearly symmetric (skewness -0.09), with a bell-shaped histogram and a *qq* plot close to the line through the middle of the distribution. ■

Exercise 1.14 (FREES exercise 1.4)**.** The file `HospitalCosts` holds a random sample of $n = 500$ Wisconsin hospital discharges of patients aged 0 to 17 from the 2003 Nationwide Inpatient Sample, with the total charge `TOTCHG`, age and sex; the aim is to model the severity of charges by age and sex. (a) Examine the distribution of `TOTCHG` by a histogram and a normal qq plot. (b) Do the same for `LNTOTCHG` $= \ln$ `TOTCHG`, and plot `LNTOTCHG` against `TOTCHG`.

Solution.

```
hosp <- read.csv("data/frees/csv/HospitalCosts.csv")
round(rbind(TOTCHG = descr(hosp$TOTCHG), LNTOTCHG = descr(log(hosp$TOTCHG))), 3)
```

```
##              n     mean   median       sd     min      max
## TOTCHG     500 2774.388 1536.500 3888.407 532.000 48388.00
## LNTOTCHG   500    7.548    7.337    0.744   6.277    10.79
##             skew
## TOTCHG     5.544
## LNTOTCHG   1.274
```



(a) The mean charge, \$2,774, is nearly twice the median, \$1,537, with a tail to \$48,388 (skewness 5.5). (b) The logarithm reduces the skewness to 1.3 but does not remove it: the transformed histogram is still skewed to the right and its *qq* plot still bends upward at the right. The plot of `LNTOTCHG` against `TOTCHG` is the concave logarithm, steep over the many small charges and flat

over the few large ones. ■

Exercise 1.15 (FREES exercise 1.5)**.** The Insurance Research Council's 2002 study of automobile injury claims: $n = 1{,}340$ closed claims from one state, with the economic loss `LOSS` in thousands, attorney involvement (`ATTORNEY`: 1 involved, 2 not) and claimant characteristics. The original file was withdrawn at the request of the survey's conductor; `AUTOBIsim`, the simulated substitute distributed in its place, is used. (a) Descriptive statistics of `LOSS`: what is the typical loss? (b) A histogram and a normal *qq* plot of `LOSS`. (c) Split the claims by attorney involvement: the typical loss in each group, a box plot by group, and a histogram and a *qq* plot for each group.

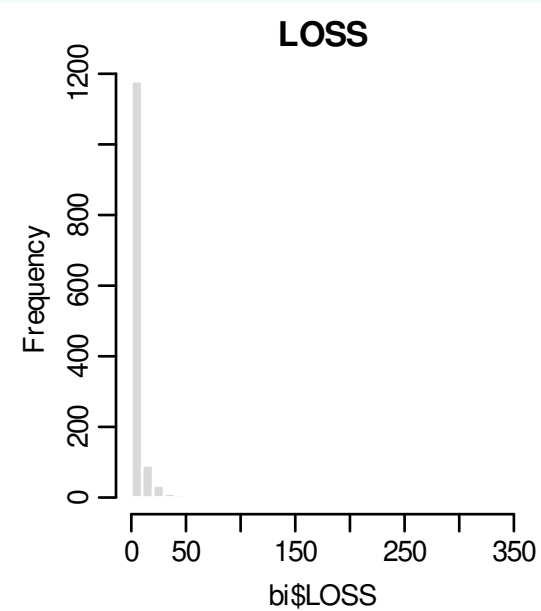
Solution.

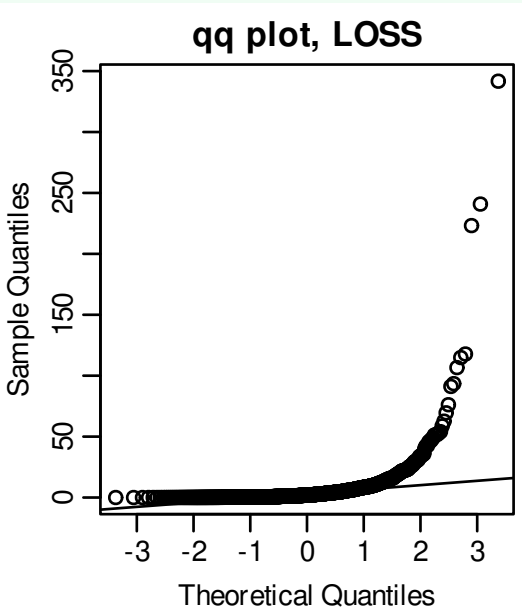
```
bi <- read.csv("data/frees/csv/AUTOBIsim.csv")
round(descr(bi$LOSS), 3)
```

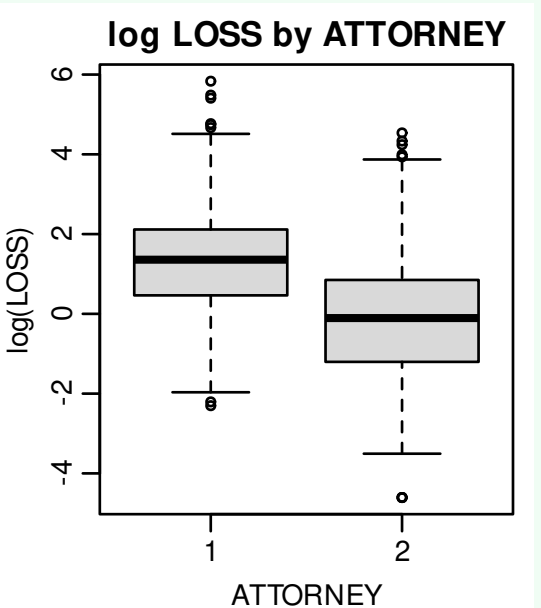
```
##         n      mean    median        sd       min       max
## 1340.000     5.682     2.000    16.174     0.010   341.720
##      skew
##    12.131
```

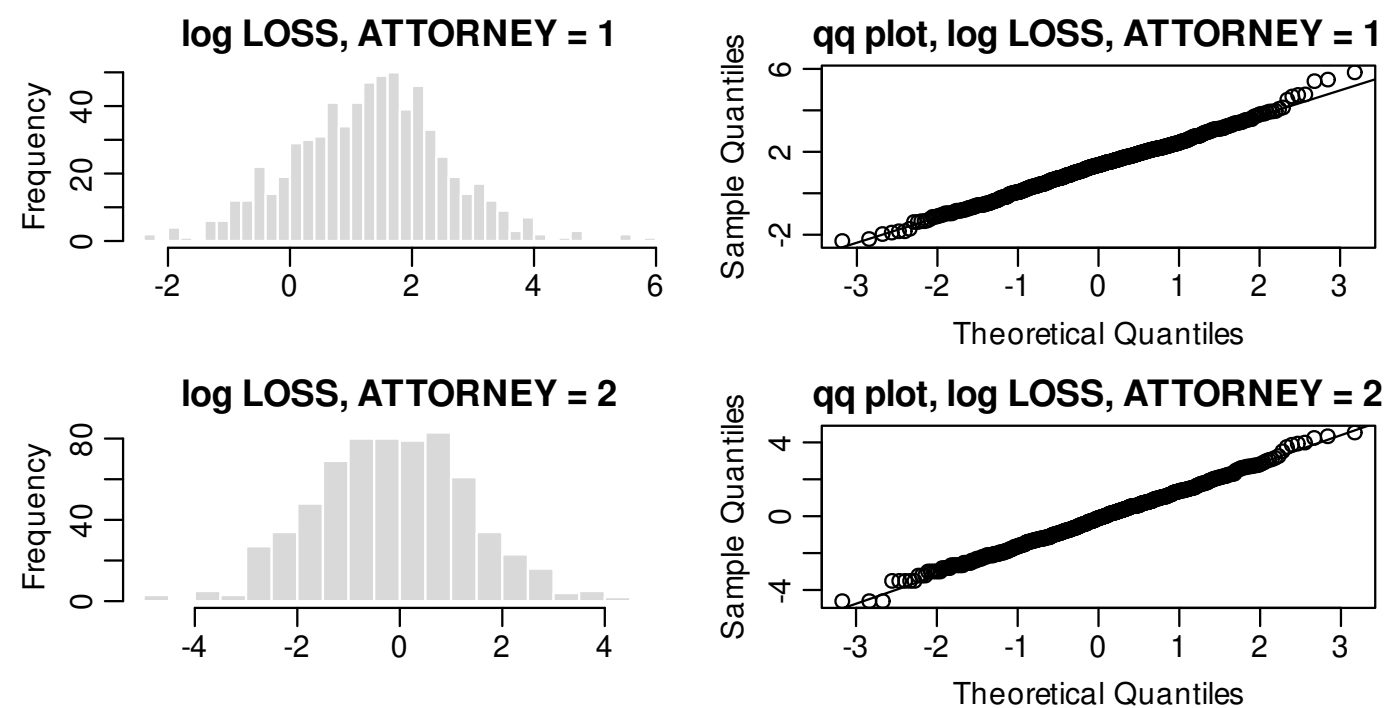
```r
round(rbind(n        = tapply(bi$LOSS, bi$ATTORNEY, length),
            mean     = tapply(bi$LOSS, bi$ATTORNEY, mean),
            median   = tapply(bi$LOSS, bi$ATTORNEY, median),
            skew     = tapply(bi$LOSS, bi$ATTORNEY, skew),
            skew_log = tapply(log(bi$LOSS), bi$ATTORNEY, skew)), 3)
```

```
##                1       2
## n        683.000 657.000
## mean       8.468   2.786
## median     3.890   0.900
## skew      10.056   7.485
## skew_log   0.052   0.081
```


LOSS
Frequency
0
200
400
600
800
1200
0
50
150
250
350
bi$LOSS


qq plot, LOSS
Sample Quantiles
0
50
150
250
350
-3
-2
-1
0
1
2
3
Theoretical Quantiles


log LOSS by ATTORNEY
log(LOSS)
-4
-2
0
2
4
6
1
2
ATTORNEY



(a) The median loss is \$2,000 and the mean \$5,682, pulled up by losses of up to \$341,720 (skewness 12.1); the median is the typical loss. (b) The histogram is a spike near zero with a long right tail, and the *qq* plot lies far above the line on the right. (c) With an attorney the median loss is \$3,890

against \$900 without, and the means are \$8,468 against \$2,786. Both groups are heavily skewed in dollars (skewness 10.1 and 7.5) and nearly symmetric in logarithms (0.05 and 0.08), so the box plot and the per-group histograms and *qq* plots are drawn on the logarithmic scale: both groups are close to normal there, with similar spreads, and the attorney group is shifted upward. ■

Exercise 1.16 (FREES exercise 1.6)**.** The file `NAICExpense` holds a sample of the property and casualty insurers in the NAIC database of 2005 annual statements, with each company's non-claim `EXPENSES` (unallocated loss adjustment, underwriting and investment expenses) in millions of dollars. (a) Examine the distribution of `EXPENSES` by a histogram and a normal *qq* plot. (b) Take the natural logarithm: has it symmetrised the distribution?

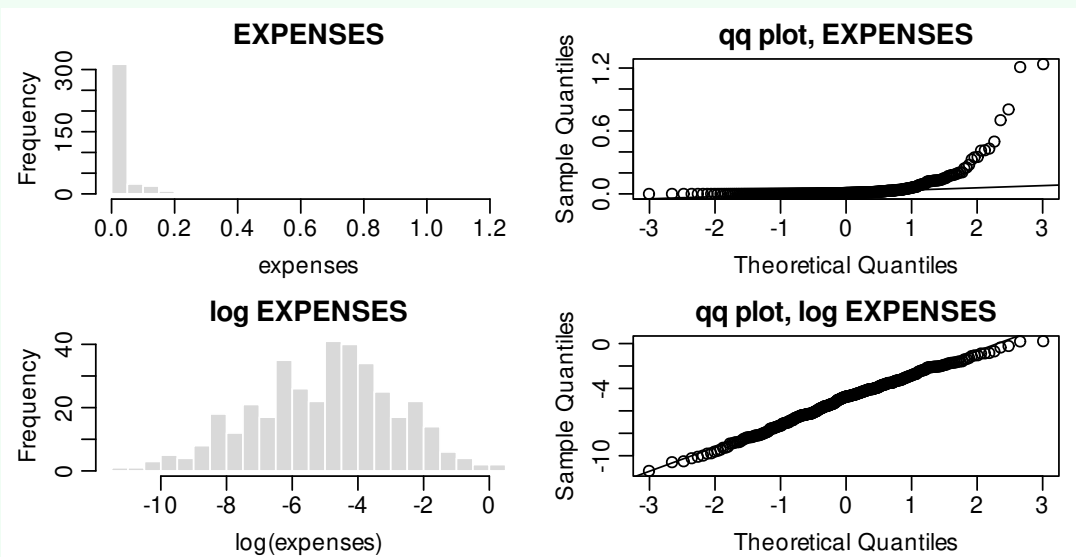
Solution. The distributed file holds 384 companies rather than the 500 of FREES's description, and four of them report non-positive expenses, which the logarithm excludes.

```
naic <- read.csv("data/frees/csv/NAICExpense.csv")
c(companies = nrow(naic), nonpositive = sum(naic$EXPENSES <= 0))
```

```
##   companies nonpositive
##         384           4
```

```
expenses <- naic$EXPENSES[naic$EXPENSES > 0]
round(rbind(EXPENSES = descr(expenses), log = descr(log(expenses))), 3)
```

```
##             n   mean median    sd    min   max   skew
## EXPENSES 380  0.044  0.009 0.122   0.00 1.237  6.481
## log      380 -5.001 -4.724 2.171 -11.35 0.213 -0.240
```



(a) The mean, 0.044, is five times the median, 0.009, and the maximum, 1.237, is ten times the

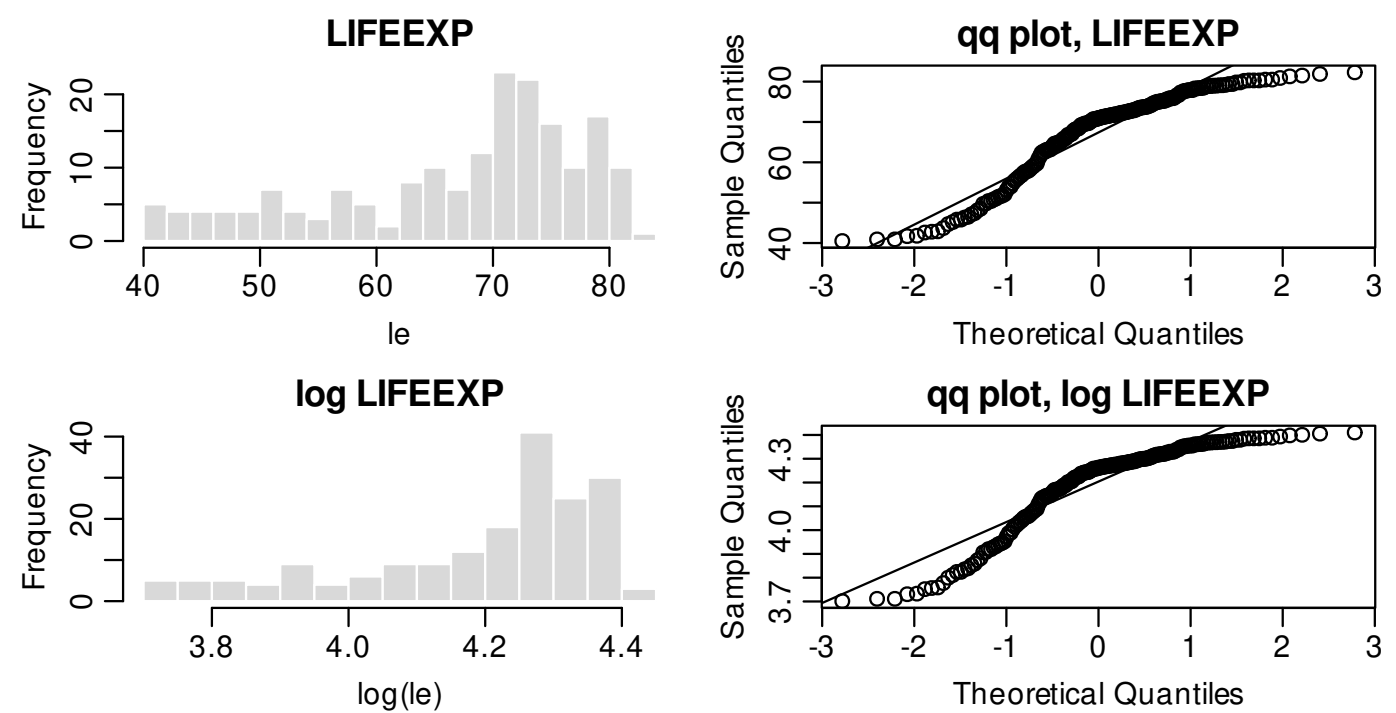
standard deviation: a spike at small values with a long right tail (skewness 6.5), and the *qq* plot bends upward at the right. (b) In logarithms the skewness is -0.24 and the *qq* plot is close to the line: the transformation has symmetrised the distribution. ■

Exercise 1.17 (FREES exercise 1.7)**.** The file `UNLifeExpectancy` holds the life expectancy at birth `LIFEEXP` of $n = 185$ countries from the UN Human Development Report, with economic and demographic characteristics (FREES Table 1.4). (a) Examine the distribution of `LIFEEXP` by a histogram and a normal *qq* plot. (b) Take the natural logarithm: has it symmetrised the distribution?

Solution.

```
un <- read.csv("data/frees/csv/UNLifeExpectancy.csv")
le <- un$LIFEEXP
round(rbind(LIFEEXP = descr(le), log = descr(log(le)), squared = descr(le^2)), 3)
```

```
##              n    mean   median       sd      min     max   skew
## LIFEEXP    185   67.05   71.000   11.082   40.500   82.30 -0.841
## log        185    4.19    4.263    0.183    3.701    4.41 -1.103
## squared    185 4617.95 5041.000 1388.197 1640.250 6773.29 -0.593
```



(a) Life expectancy is skewed to the **left**: the mean, 67.1 years, is below the median, 71.0, the skewness is -0.84, and the *qq* plot falls below the line on the left, where the countries with life expectancies in the forties and fifties form the lower tail. (b) The logarithm stretches the lower

tail further and moves the skewness to -1.10: it symmetrises right skew only. A power above one works in the other direction; the square has skewness -0.59. ■

Exercise 1.18 (SOA sample question 12)**.** Determine which of the following statements is true. (A) Linear regression is a flexible approach. (B) Lasso is more flexible than a linear regression approach. (C) Bagging is a low flexibility approach. (D) There are methods that have high flexibility and are also easy to interpret. (E) None of (A), (B), (C) or (D) is true.

Solution. Flexibility is the range of shapes a method can produce for $\hat{f}$ (Definition 1.7), and the ordering of the methods is the one ISLR Figure 2.7 records.

(A) is false: least squares produces only the functions (5), a $(p+1)$-parameter family, and is the least flexible method on the list after subset selection and the lasso.

(B) is false. The lasso fits the same linear form (5) but by a procedure that shrinks the coefficients and sets some of them exactly to zero, so every lasso fit is a linear fit while some linear fits are unavailable to it; its range of shapes is smaller and it is **less** flexible, and more interpretable, than least squares.

(C) is false: bagging averages many trees, each a step function with arbitrarily many pieces, and sits at the flexible end of the ordering with boosting, support vector machines and neural networks.
(D) is false in the sense intended: in ISLR Figure 2.7, which records the ordering of Definition 1.7, no method sits in the corner of high flexibility and high interpretability; there is a trade-off between the two.
Hence (E): none of the four statements is true. ■

Exercise 1.19 (SOA sample question 50)**.** Determine which of the following statements regarding statistical learning methods is or are true. (I) Methods that are highly interpretable are more likely to be highly flexible. (II) When inference is the goal, there are clear advantages to using a lasso method versus a bagging method. (III) Using a more flexible method will produce a more accurate prediction against unseen data.

Solution. (I) is false: it reverses Definition 1.7, under which interpretability falls as flexibility rises; a ninth-degree polynomial is harder to read than a line.
(II) is true. Inference (Definition 1.3) asks which predictors matter and with what sign and size. A lasso fit is the linear form (5) with most coefficients exactly zero, from which those answers can

be read; a bagged model is an average of hundreds of trees from which no such statement can be extracted. The comparison concerns what the fitted object says, not which predicts better.
(III) is false. By Proposition 1.15 the expected test MSE is squared bias plus variance plus $\operatorname{var}(\varepsilon)$; added flexibility lowers the bias and raises the variance, and once the bias is small the sum rises. The simulation of §2.1 shows it: the test MSE of the polynomial fits falls to a minimum and then rises with the degree while the training MSE keeps falling.
So statement II only is true. ■

Exercise 1.20 (ISLR §5.4, exercise 3)**.** (a) Explain how k-fold cross-validation is implemented. (b) State its advantages and disadvantages relative to (i) the validation set approach and (ii) LOOCV.

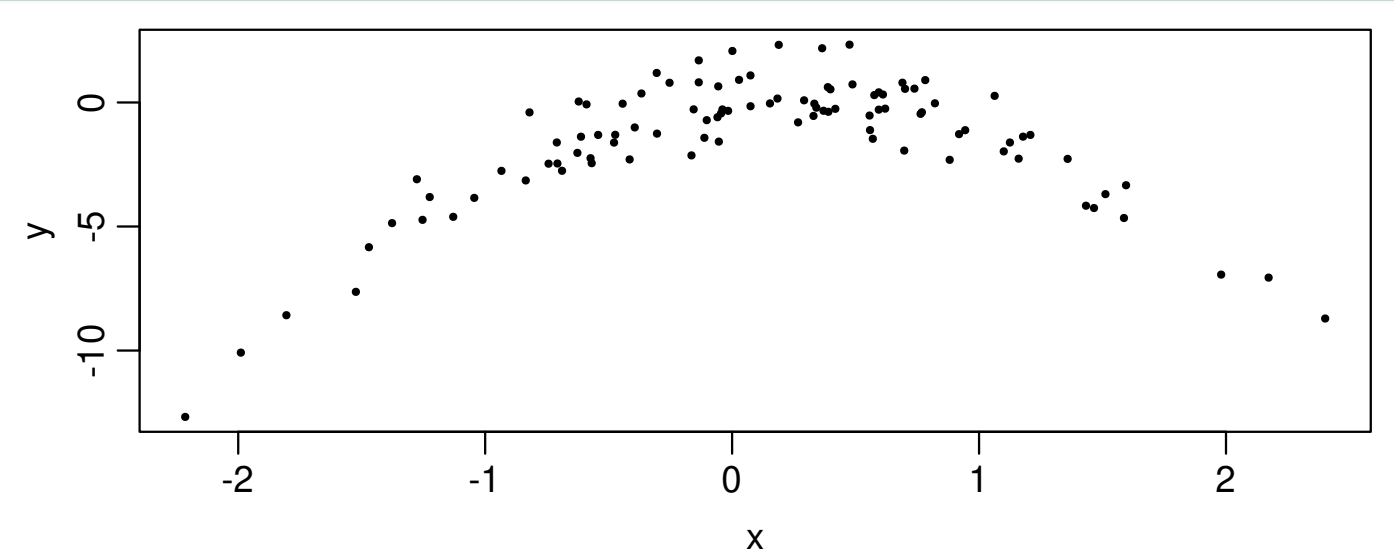
Solution. (a) Definition 1.25: partition the observations at random into k folds of about n/k observations; for $j = 1, \dots, k$ fit on the other $k-1$ folds and compute the error on fold j; average the k errors.
(b)(i) Against the validation set approach: each observation is used for training in $k-1$ of the fits and for validation once, so the estimate is far less variable across random splits, and each fit uses

$(k-1)n/k$ rather than $n/2$ observations, so the upward bias is smaller. The cost is k fits instead of one.
(ii) Against LOOCV: k-fold costs k fits instead of n (except for least squares, where (12) makes LOOCV one fit); by ISLR's argument its estimate has lower variance, since the k fits overlap less than the n nearly identical LOOCV fits, and Proposition 1.26 shows that a smaller ρ lowers the variance of an average with σ^2 and k fixed (a heuristic only: the two estimators average quantities with different σ^2 and k, and for least squares the ordering can reverse); and it has somewhat higher bias, each fit using $(k-1)n/k$ rather than $n-1$ observations. It also depends on the random partition, whereas LOOCV does not. ■

Exercise 1.21 (ISLR §5.4, exercise 8)**.** Generate $n = 100$ observations by `x <- rnorm(100); y <- x - 2*x^2 + rnorm(100)` after `set.seed(1)`. (a) What are n and p, and what is the model in equation form? (b) Plot y against x. (c) Compute the LOOCV errors of the least squares fits of degrees 1 to 4. (d) Repeat with another seed and explain. (e) Which model has the smallest LOOCV error, and is that expected? (f) Compare with the significance of the coefficients in the degree-4 fit.

Solution. (a) $n = 100$ and $p = 1$; the model is $Y = X - 2X^2 + \varepsilon$ with X and ε standard normal, a quadratic truth.

```
set.seed(1)
xs <- rnorm(100); ys <- xs - 2*xs^2 + rnorm(100)
ds <- data.frame(x = xs, y = ys)
```



(b) The scatter is an inverted parabola with noise: a straight line cannot fit it.

```r
loocv <- function(seed) { set.seed(seed)
  sapply(1:4, function(dg) cv.glm(ds, glm(y ~ poly(x, dg), data = ds))$delta[1]) }
rbind(degree = 1:4, seed_1 = round(loocv(1), 4), seed_99 = round(loocv(99), 4))
```

```
##           [,1]   [,2]   [,3]   [,4]
## degree   1.000 2.0000 3.0000 4.0000
## seed_1   7.288 0.9374 0.9566 0.9539
## seed_99  7.288 0.9374 0.9566 0.9539
```

```r
round(coef(summary(lm(y ~ poly(x, 4), data = ds))), 4)
```

```
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept)   -1.5500     0.0959 -16.1620   0.0000
## poly(x, 4)1    6.1888     0.9591   6.4531   0.0000
## poly(x, 4)2  -23.9483     0.9591 -24.9708   0.0000
## poly(x, 4)3    0.2641     0.9591   0.2754   0.7836
## poly(x, 4)4    1.2571     0.9591   1.3108   0.1931
```

(c), (d) The two rows are identical: LOOCV involves no random split, the n training sets being fixed by the data, so the seed cannot matter. (e) Degree 2 has the smallest error, as expected since the truth is quadratic; degrees 3 and 4 add variance and remove no bias, and their errors are slightly

larger. (f) The linear and quadratic coefficients are overwhelmingly significant and the cubic and quartic are not, which agrees with the cross-validation. The agreement is not automatic: the t tests judge coefficients on the training data under the model's assumptions, while cross-validation estimates prediction error on held-out data without them. ■

Exercise 1.22 (ISLR §5.4, exercise 1; beyond the syllabus)**.** Two assets have returns X and Y with variances σ_X^2, σ_Y^2 and covariance σ_{XY}. Show that $\operatorname{var}(\alpha X + (1-\alpha)Y)$ is minimised by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}},$$

then estimate α on `ISLR2::Portfolio` and give its bootstrap standard error.

Solution. By bilinearity of the covariance,

$$v(\alpha) = \operatorname{var}(\alpha X + (1-\alpha)Y) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}.$$

Differentiate in α:

$$v'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 2\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) - 2(\sigma_Y^2 - \sigma_{XY}).$$

Setting $v'(\alpha) = 0$ gives the stated α. It is a minimum because $v''(\alpha) = 2(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = 2\,\mathrm{var}(X - Y) > 0$ unless $X - Y$ is constant; the same quantity is the denominator, so α is well defined exactly when v is not flat.

```
alpha.fn <- function(data, index) {
  X <- data$X[index]; Y <- data$Y[index]
  (var(Y) - cov(X, Y)) / (var(X) + var(Y) - 2 * cov(X, Y))
}
alpha.fn(Portfolio, 1:100)
```

```
## [1] 0.5758
```

```
set.seed(7)
ba <- boot(Portfolio, alpha.fn, R = 1000)
c(alpha_hat = ba$t0, boot_SE = sd(ba$t))
```

```
## alpha_hat   boot_SE
##   0.57583   0.08978
```

The plug-in estimate replaces the three population moments by sample moments. It is a ratio of estimated variances and covariances, for which no elementary standard-error formula exists at finite n, which is the case the bootstrap is for. ■

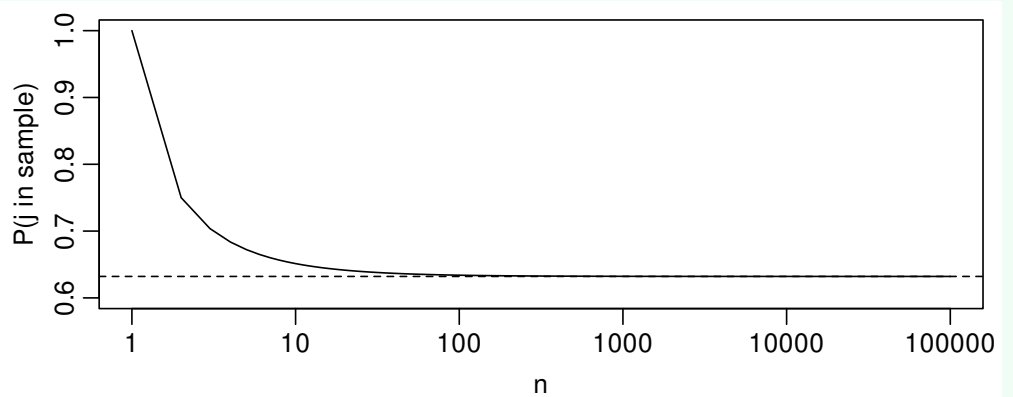
Exercise 1.23 (ISLR §5.4, exercise 2; beyond the syllabus)**.** (a) What is the probability that the first bootstrap observation is not the j-th observation of the original sample? (b) The second? (c) Argue that the probability that the j-th observation is not in the bootstrap sample is $(1 - 1/n)^n$. (d), (e), (f) Compute the probability that it **is** in the sample for $n = 5$, 100 and 10,000. (g) Plot that probability against n. (h) Estimate it by simulation for $n = 100$, $j = 4$.

Solution. (a) Each bootstrap draw is uniform on the n observations, so it equals observation j with probability $1/n$ and is not j with probability $1 - 1/n$. (b) The same, the draws being identically distributed. (c) The n draws are independent, so the probability that none of them is j is the product of n factors $1 - 1/n$, which is Step 1 of Proposition 1.29.

```
nn <- c(5, 100, 10000)
rbind(n = nn, P_in = 1 - (1 - 1/nn)^nn, limit = 1 - exp(-1))     # (d), (e), (f)
```

```
##          [,1]     [,2]       [,3]
## n      5.0000 100.0000 10000.0000
## P_in   0.6723   0.6340     0.6321
## limit  0.6321   0.6321     0.6321
```

```r
op <- par(mar = c(3.2, 3.2, 0.6, 0.6), mgp = c(1.9, 0.6, 0))
n_grid <- 1:100000
plot(n_grid, 1 - (1 - 1/n_grid)^n_grid, log = "x", type = "l", ylim = c(0.6, 1),
     xlab = "n", ylab = "P(j in sample)"); abline(h = 1 - exp(-1), lty = 2)
par(op)
```




```r
set.seed(1)
store <- replicate(10000, sum(sample(1:100, replace = TRUE) == 4) > 0)
mean(store)                                                    # (h)
```

```
## [1] 0.6417
```

(g) The curve falls steeply from 1 at $n = 1$ and is within a few thousandths of the limit $1 - e^{-1}$ by $n = 100$: the convergence is of order $1/n$, by the series in the proof of Proposition 1.29. (h) The simulated frequency differs from $1 - (1 - 1/100)^{100} = 0.634$ by less than two simulation standard errors, $\sqrt{p(1-p)/10{,}000} \approx 0.005$. ■

Exercise 1.24 (ISLR §5.4, exercise 4; beyond the syllabus)**.** A statistical learning method is used to predict Y at a particular value of X. Describe how the standard deviation of that prediction can be estimated.

Solution. Draw B bootstrap samples of the training data (Definition 1.28); fit the method to each, giving $\hat{f}^{*b}$; evaluate each at the given X, giving B predictions $\hat{f}^{*b}(X)$; and report their standard deviation (17). This estimates the standard deviation of $\hat{f}(X)$ over the draw of the training set, the square root of the variance term in Proposition 1.15. It does not include $\mathrm{var}(\varepsilon)$, the spread of a new Y about $f(X)$, which no resampling of the training data can measure. ■

Exercise 1.25 (ISLR §5.4, exercise 9; beyond the syllabus)**.** For `medv` in `Boston`: (a) estimate the population mean μ; (b) give the standard error of $\hat{\mu}$ by the formula $s/\sqrt{n}$; (c) estimate it

by the bootstrap and compare; (d) give a 95% confidence interval $\hat{\mu} \pm 2\text{SE}(\hat{\mu})$ and compare with `t.test`; (e), (f) estimate the population median and its standard error by the bootstrap; (g), (h) the same for the tenth percentile.

Solution.

```r
mv <- Boston$medv
set.seed(3)
b_mean <- boot(mv, function(v, i) mean(v[i]), R = 2000)
b_med  <- boot(mv, function(v, i) median(v[i]), R = 2000)
b_q10  <- boot(mv, function(v, i) quantile(v[i], 0.1), R = 2000)
rbind(estimate = c(mean = mean(mv), median = median(mv), q10 = unname(quantile(mv,
 ↪  0.1))),
      boot_SE  = c(sd(b_mean$t), sd(b_med$t), sd(b_q10$t)),
      formula  = c(sd(mv)/sqrt(length(mv)), NA, NA))
```

```
##              mean  median    q10
## estimate 22.5328 21.2000 12.750
## boot_SE   0.4139  0.3773  0.504
## formula   0.4089      NA     NA
```

```
rbind(boot_CI = mean(mv) + c(-2, 2) * sd(b_mean$t),
      t_test  = as.numeric(t.test(mv)$conf.int))
```

```
##           [,1]  [,2]
## boot_CI 21.71 23.36
## t_test  21.73 23.34
```

(a)–(c) The bootstrap standard error of the mean agrees with $s/\sqrt{n}$ to about one per cent. That is expected: a bootstrap sample is n independent draws from the empirical distribution, whose variance is $\hat{\sigma}^2 = n^{-1}\sum_i (x_i - \bar{x})^2$, so the exact bootstrap standard deviation of the mean is $\hat{\sigma}/\sqrt{n} = \sqrt{(n-1)/n}\, s/\sqrt{n}$, and $\sqrt{505/506} = 0.999$; the rest is Monte-Carlo error at $B = 2000$. (d) The two intervals agree to a few hundredths: the bootstrap interval uses the multiplier 2, the t interval $t_{505,0.975} = 1.965$. (e)–(h) The median and the tenth percentile are not smooth functions of the sample, and their standard errors involve the unknown population density at those quantiles; the bootstrap needs neither. The tenth percentile has the largest standard error of the three, the data being sparser in the lower tail than at the centre. ■